

CONTINUING AIRWORTHINESS

(IR + AMC & GM)



Easy Access Rules for Continuing Airworthiness (Regulation (EU) No 1321/2014)

EASA eRULES

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¹ The published date represents the date when the consolidated version of the EAR book was generated.

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This document is issued by the European Union Aviation Safety Agency (referred to as both 'EASA' and 'the Agency') to provide its stakeholders with an updated, consolidated, and easy-to-read publication. It has been prepared by putting together the officially published EU regulations with the related EASA acceptable means of compliance (AMC) and guidance material (GM) (including their amendments) adopted so far. However, this document is not an official publication, and EASA accepts no liability for damage of any kind resulting from the risks inherent in its use.

LIST OF REVISIONS

Published	Reason for revision
January 2017	First Easy Access Rules document powered by eRules.
June 2017	To incorporate ED Decision 2017/016/R amending Appendix I to AMC to Part-66 'Aircraft type ratings for Part-66 aircraft maintenance licences'.
April 2019	To incorporate amending Regulation (EU) 2018/1142 , and ED Decision 2019/009/R introducing AMC/GM to the cover regulation and amending AMC/GM to Part-M, Part-145, Part-66, Part-147 and Part-T.
June 2020	To incorporate the following amending regulations: – Regulation (EU) 2019/1383 introducing in particular Part-ML (Annex Vb), Part-CAMO (Annex Vc) and Part-CAO (Annex Vd); – Corrigendum to Regulation (EU) 2019/1383 – Regulation (EU) 2019/1384 – Regulation (EU) 2020/270 To incorporate the following ED Decisions: – ED Decision 2019/024/R amending Appendix I to AMC to Part-66 'Aircraft type ratings for Part-66 aircraft maintenance licences'. – ED Decision 2020/002/R amending AMC/GM to Part-M, Part-145, Part-66, Part-147, Part-T and to the cover regulation and introducing AMC/GM to Part-ML, Part-CAMO and Part-CAO
February 2021	To incorporate amending Regulation (EU) 2020/1159 as regards the introduction of new additional airworthiness requirements. To incorporate the related ED Decision 2020/023/R.
July 2021	To incorporate amending Regulation (EU) 2021/700 as regards the maintenance data and the installation of certain aircraft components during maintenance. To incorporate the related ED Decision 2021/009/R.
September 2022	To incorporate amending Regulation (EU) 2022/410 as regards the continuing airworthiness management in a single air carrier business grouping. To incorporate the related ED Decision 2022/017/R. To withdraw the provisions no longer applicable after 18 May 2022 pursuant to Regulation (EU) 2021/700 and related ED Decision 2021/009/R.
October 2022 (Advance revision)	To incorporate amending Regulation (EU) 2021/1963 as regards safety management systems in maintenance organisations. To incorporate the related ED Decision 2022/011/R.
December 2022	To incorporate Corrigendum to Regulation (EU) 2019/1383 (OJ L 297/83 of 17.11.2022). To delete erroneous content (points 5 and 6 wrongly incorporated) in AMC1 145.A.30(d) of the advance revision from October 2022.

Published	Reason for revision
	To incorporate formatting changes and withdraw the markings "Applicable from 2 December 2022", which were introduced by the advance revision and become unneeded as this revision is published on the same date.
July 2024	To incorporate the following amending regulations and related ED Decisions: – Regulation (EU) 2022/1360 and ED Decision 2023/013/R, as regards the implementation of more proportionate requirements for aircraft used for sport and recreational aviation; – Regulation (EU) 2023/203 and ED Decision 2023/010/R, as regards requirements for the management of information security risks with a potential impact on aviation safety for organisations; – Regulation (EU) 2023/989, ED Decision 2023/019/R, and Corrigendum to ED Decision 2023/019/R, as regards the review of Part-66 and the introduction of new training methods and new teaching technologies.

NOTE FROM THE EDITOR

The content of this document is arranged as follows: the cover regulation (recitals and articles) of the implementing rule (IR) appear first, then the IR annex points, followed by the related acceptable means of compliance (AMC) and guidance material (GM).

All elements (i.e. articles, IRs, AMC, and GM) are colour-coded and can be identified according to the illustration below. The EU regulation or EASA Executive Director (ED) decision through which the article, IR, AMC, or GM was introduced or last amended is indicated below the article, IR, AMC, or GM title in *italics*.

<u>Cover regulation article</u>	<i>Regulation</i>
Implementing rule	<i>Regulation</i>
Acceptable means of compliance	<i>ED Decision</i>
Guidance material	<i>ED Decision</i>

Note 1: In some instances (e.g. [145.A.30](#) Personnel requirements), the source is indicated at the level of point paragraph (e.g. 145.A.30(a)).

This document will be updated regularly to incorporate further amendments.

The format of this document has been adjusted to make it user-friendly and for reference purposes. Any comments should be sent to erules@easa.europa.eu.

INCORPORATED AMENDMENTS

IMPLEMENTING RULES (IRs) (COMMISSION REGULATIONS)

Incorporated Commission Regulation	Affected Part	Regulation amendment	Applicability date ¹
Regulation (EU) No 1321/2014	Annex I (Part-M)	Recast	06/01/2015
	Annex II (Part-145)		
	Annex III (Part-66)		
	Annex IV (Part-147)		
Regulation (EU) 2015/1088	Annex I (Part-M)	First amending regulation	27/07/2015
	Annex II (Part-145)		
	Annex IV (Part-147)		
Regulation (EU) 2015/1536	Annex I (Part-M)	Second amending regulation	25/08/2016
	Annex II (Part-145)		
	Annex III (Part-66)		
	Annex Va (Part-T)		
Regulation (EU) 2018/1142	Annex I (Part-M)	Third amending regulation	05/03/2019* * Refer to Article 8 of the cover regulation
	Annex II (Part-145)		
	Annex III (Part-66)		
	Annex IV (Part-147)		
	Annex Va (Part-T)		
Regulation (EU) 2019/1383 and Corrigendum to Regulation (EU) 2019/1383 (OJ L 230/7 of 6.9.2019) and Corrigendum to Regulation (EU) 2019/1383 (OJ L 297/83 of 17.11.2022)	Annex I (Part-M)	Fourth amending regulation	24/03/2020
	Annex II (Part-145)		
	Annex III (Part-66)		
	Annex IV (Part-147)		
	Annex Va (Part-T)		
	Annex Vb (Part-ML)		
	Annex Vc (Part-CAMO)		
	Annex Vd (Part-CAO)		
Regulation (EU) 2019/1384	Annex I (Part-M)	Fifth amending regulation	24/03/2020
	Annex Vb (Part-ML)		
Regulation (EU) 2020/270	Annex I (Part-M)	Sixth amending regulation	24/03/2020
	Annex II (Part-145)		
	Annex III (Part-66)		
	Annex IV (Part-147)		
	Annex Vb (Part-ML)		
	Annex Vc (Part-CAMO)		
	Annex Vd (Part-CAO)		

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¹ This is the date of application (i.e. the date from which an act or a provision in an act produces its full legal effects) as defined in the relevant cover regulation article. Some provisions of the regulations though may be applicable at a different date (e.g. deferred applicability) as specified in the relevant article of the regulation. Besides, there may be some opt-outs (derogations from certain provisions) notified by the Member States.

Incorporated Commission Regulation	Affected Part	Regulation amendment	Applicability date
Regulation (EU) 2020/1159	Annex I (Part-M)	Seventh amending regulation	26/02/2021
Regulation (EU) 2021/700	Annex I (Part-M)	Eighth amending regulation	18/05/2021
	Annex II (Part-145)		18/05/2022
	Annex III (Part-66)		18/05/2022
	Annex Vb (Part-ML)		18/05/2021
	Annex Vd (Part-CAO)		18/05/2022
Regulation (EU) 2022/410	Annex I (Part-M)	Ninth amending regulation	18/05/2021
	Annex Vc (Part-CAMO)		18/05/2021
Regulation (EU) 2021/1963	Annex I (Part-M)	Tenth amending regulation	31/03/2022
	Annex II (Part-145)		02/12/2021
	Annex Vb (Part-ML)		02/12/2022
Regulation (EU) 2022/1360	Annex I (Part-M)	Eleventh amending regulation	02/12/2021
	Annex II (Part-145)		02/12/2022
	Annex III (Part-66)		02/12/2022
	Annex Vb (Part-ML)		02/12/2021
	Annex Vc (Part-CAMO)		02/12/2021
Regulation (EU) 2023/989	Annex I (Part-M)	Twelfth amending regulation	25/08/2023
	Annex III (Part-66)		12/06/2023
	Annex IV (Part-147)		12/06/2024
	Annex Vb (Part-ML)		12/06/2024
Regulation (EU) 2023/203	Annex II (Part-145)	Thirteenth amending regulation	12/06/2023
	Annex III (Part-66)		22/02/2026
	Annex Vc (Part-CAMO)		22/02/2026

Notes:

To access the official versions, please click on the hyperlinks provided above.

Amending regulations issued to correct language versions other than English (e.g. Regulation (EU) 2017/334) are not considered in this document.

AMC/GM TO IRs (ED DECISIONS)

Incorporated ED Decision	Affected AMC/GM	AMC/GM Issue No, Amendment No	Applicability date
ED Decision 2015/029/R	AMC/GM to Annex I (Part-M)	Issue 2 (Recast)	18/12/2015
	AMC/GM to Annex II (Part-145)		
	AMC/GM to Annex III (Part-66)		
	AMC/GM to Annex IV (Part-147)		
ED Decision 2016/011/R	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 1	25/08/2016
	AMC/GM to Annex II (Part-145)		
	AMC/GM to Annex III (Part-66)		
	AMC/GM to Annex Va (Part-T)	Issue 1	
ED Decision 2017/016/R	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 2	02/06/2017
ED Decision 2019/009/R	AMC/GM to articles	Issue 1	05/03/2019
	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 2	
	AMC/GM to Annex II (Part-145)		
	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 3	
	AMC/GM to Annex IV (Part-147)	Issue 2, Amendment 1	
	AMC/GM to Annex Va (Part-T)	Issue 1, Amendment 1	
ED Decision 2019/024/R	AMC & GM to Annex III (Part-66)	Issue 2, Amendment 4	19/11/2019
ED Decision 2020/002/R	AMC/GM to articles	Issue 1, Amendment 1	24/03/2020
	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 3	
	AMC/GM to Annex II (Part-145)	Issue 2, Amendment 3	
	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 5	
	AMC/GM to Annex IV (Part-147)	Issue 2, Amendment 2	
	AMC/GM to Annex Va (Part-T)	Issue 1, Amendment 2	
	AMC/GM to Annex Vb (Part-ML)	Issue 1	
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1	
	AMC/GM to Annex Vd (Part-CAO)	Issue 1	
ED Decision 2020/023/R	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 4	26/02/2021
ED Decision 2021/009/R	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 5	16/06/2021
			18/05/2022
	AMC/GM to Annex II (Part-145)	Issue 2, Amendment 4	16/06/2021
			18/05/2022
	AMC/GM to Annex Va (Part-T)	Issue 1, Amendment 3	16/06/2021
	AMC/GM to Annex Vb (Part-ML)	Issue 1, Amendment 1	16/06/2021
		18/05/2022	
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1, Amendment 1	16/06/2021
	AMC/GM to Annex Vd (Part-CAO)	Issue 1, Amendment 1	16/06/2021
ED Decision 2022/017/R	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 7	03/09/2022
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1, Amendment 3	03/09/2022

Incorporated ED Decision	Affected AMC/GM	AMC/GM Issue No, Amendment No	Applicability date
ED Decision 2022/011/R and Corrigendum to ED Decision 2022/011/R	AMC/GM to articles	Issue 1, Amendment 2	02/12/2022
	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 6	02/12/2022
	AMC/GM to Annex II (Part-145)	Issue 2, Amendment 5	02/12/2022
	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 6	02/12/2022
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1, Amendment 2	11/05/2022 02/12/2022
	AMC/GM to Annex Vd (Part-CAO)	Issue 1, Amendment 2	02/12/2022
ED Decision 2023/013/R	AMC/GM to Annex I (Part-M)	Issue 2, Amendment 8	21/10/2023
	AMC/GM to Annex II (Part-145)	Issue 2, Amendment 7	
	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 7	
	AMC/GM to Annex Vb (Part-ML)	Issue 1, Amendment 2	
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1, Amendment 5	
	AMC/GM to Annex Vd (Part-CAO)	Issue 1, Amendment 3	
ED Decision 2023/019/R and Corrigendum to ED Decision 2023/019/R	AMC/GM to articles	Issue 1, Amendment 3	12/06/2024
	AMC/GM to Annex III (Part-66)	Issue 2, Amendment 8	
	AMC/GM to Annex IV (Part-147)	Issue 2, Amendment 3	
	AMC/GM to Annex Vd (Part-CAO)	Issue 1, Amendment 4	
ED Decision 2023/010/R	AMC/GM to Annex II (Part-145)	Issue 2, Amendment 6	22/02/2026
	AMC/GM to Annex Vc (Part-CAMO)	Issue 1, Amendment 4	

Note:

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APPENDICES TO ANNEX III (PART-66)

Appendix I — Basic Knowledge Requirements (except for category L licence)

1. Knowledge levels for Category A, B1, B2, B2L, B3 and C aircraft maintenance licences

Regulation (EU) 2018/1142

Basic knowledge for categories A, B1, B2, B2L and B3 is indicated by knowledge levels (1, 2 or 3) of each applicable subject. Category C applicants shall meet either the category B1 or the category B2 basic knowledge levels.

The knowledge level indicators are defined on 3 levels as follows:

- *LEVEL 1: A familiarisation with the principal elements of the subject.*

Objectives:

- (a) The applicant should be familiar with the basic elements of the subject.
- (b) The applicant should be able to give a simple description of the whole subject, using common words and examples.
- (c) The applicant should be able to use typical terms.

- *LEVEL 2: A general knowledge of the theoretical and practical aspects of the subject and an ability to apply that knowledge.*

Objectives:

- (a) The applicant should be able to understand the theoretical fundamentals of the subject.
- (b) The applicant should be able to give a general description of the subject using, as appropriate, typical examples.
- (c) The applicant should be able to use mathematical formulae in conjunction with physical laws describing the subject.
- (d) The applicant should be able to read and understand sketches, drawings and schematics describing the subject.
- (e) The applicant should be able to apply his knowledge in a practical manner using detailed procedures.

- *LEVEL 3: A detailed knowledge of the theoretical and practical aspects of the subject and a capacity to combine and apply the separate elements of knowledge in a logical and comprehensive manner.*

Objectives:

- (a) The applicant should know the theory of the subject and interrelationships with other subjects.
- (b) The applicant should be able to give a detailed description of the subject using theoretical fundamentals and specific examples.
- (c) The applicant should understand and be able to use mathematical formulae related to the subject.

- (d) The applicant should be able to read, understand and prepare sketches, simple drawings and schematics describing the subject.
- (e) The applicant should be able to apply his knowledge in a practical manner using manufacturer's instructions.
- (f) The applicant should be able to interpret results from various sources and measurements and apply corrective action where appropriate.

2. Modularisation

Regulation (EU) 2023/989

Qualification on basic subjects for each aircraft maintenance licence category or subcategory shall be in accordance with the following matrix, where applicable subjects are indicated by an 'X', while 'n/a' means that the subject module is neither applicable nor required.

Subject module	B1.1 A1	B1.2 A2	B1.3 A3	B1.4 A4	B3	B2	B2L	C
	Turbine engine	Piston engine	Turbine engine	Piston engine	Piston-engine non-pressurised aeroplanes MTOM ≤ 2 t			
1. MATHEMATICS	X	X	X	X	X	X	X	X
2. PHYSICS	X	X	X	X	X	X	X	X
3. ELECTRICAL FUNDAMENTALS	X	X	X	X	X	X	X	X
4. ELECTRONICS FUNDAMENTALS	X (n/a for A1)	X (n/a for A2)	X (n/a for A3)	X (n/a for A4)	X	X	X	X
5. DIGITAL TECHNIQUES/ELECTRONIC INSTRUMENT SYSTEMS	X	X	X	X	X	X	X	X
6. MATERIALS AND HARDWARE	X	X	X	X	X	X	X	X
7. MAINTENANCE PRACTICES	X	X	X	X	X	X	X	X
8. BASIC AERODYNAMICS	X	X	X	X	X	X	X	X
9. HUMAN FACTORS	X	X	X	X	X	X	X	X
10. AVIATION LEGISLATION	X	X	X	X	X	X	X	X
11. AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	X	X	n/a	n/a	X	n/a	n/a	11, 15 & 17 as B1.1

Subject module	B1.1 A1	B1.2 A2	B1.3 A3	B1.4 A4	B3	B2	B2L	C
	Turbine engine	Piston engine	Turbine engine	Piston engine	Piston-engine non-pressurised aeroplanes MTOM ≤ 2 t			
12. HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	n/a	n/a	X	X	n/a	n/a	n/a	or 11, 16 & 17 as B1.2 or
13. AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	n/a	n/a	n/a	n/a	n/a	X	X	12 & 15 as B1.3 or
14. PROPULSION	n/a	n/a	n/a	n/a	n/a	X	X	12 & 16 as B1.4 or
15. GAS TURBINE ENGINES	X	n/a	X	n/a	n/a	n/a	n/a	13 & 14 as B2
16. PISTON ENGINE	n/a	X	n/a	X	X	n/a	n/a	
17. PROPELLER	X	X	n/a	n/a	X	n/a	n/a	

MODULE 1. MATHEMATICS (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 1. MATHEMATICS	LEVEL	
	A	B1 B2 B2L B3
1.1 Arithmetic	1	2
1.2 Algebra		
(a) Simple algebraic expressions;	1	2
(b) Equations.	—	1
1.3 Geometry		
(a) Simple geometrical constructions;	—	1
(b) Graphical representation;	2	2
(c) Trigonometry.	—	2

MODULE 2. PHYSICS (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 2. PHYSICS	LEVEL	
	A B3	B1 B2 B2L
2.1 Matter	1	2
2.2. Mechanics		
2.2.1 Statics	1	2
2.2.2 Kinetics	1	2
2.2.3 Dynamics		
(a) Mass, force and energy;	1	2
(b) Momentum and conservation of momentum.	1	2
2.2.4 Fluid dynamics		
(a) Gravity and density;	2	2
(b) Viscosity; compressibility on fluids; static, dynamic, and total pressure.	1	2
2.3 Thermodynamics		
(a) Temperature;	2	2
(b) Heat.	1	2
2.4 Optics (light)	—	2
2.5 Wave motion and sound	—	2

MODULE 3. ELECTRICAL FUNDAMENTALS (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 3. ELECTRICAL FUNDAMENTALS	LEVEL		
	A	B1 B2 B2L	B3
3.1 Electron theory	1	1	1
3.2 Static electricity and conduction	1	2	1
3.3 Electrical terminology	1	2	1
3.4 Generation of electricity	1	1	1
3.5 Sources of DC electricity	1	2	2
3.6 DC circuits	1	2	1
3.7 Resistance/resistor			
(a) Resistance;	—	2	1
(b) Resistors.	—	1	—
3.8 Power	—	2	1
3.9 Capacitance/capacitor	—	2	1
3.10 Magnetism			
(a) Theory of magnetism;	—	2	1
(b) Magnetomotive force.	—	2	1

MODULE 3. ELECTRICAL FUNDAMENTALS	LEVEL		
	A	B1 B2 B2L	B3
3.11 Inductance/inductor	—	2	1
3.12 DC motor/generator theory	—	2	1
3.13 AC theory	1	2	1
3.14 Resistive (R), capacitive (C) and inductive (L) circuits	—	2	1
3.15 Transformers	—	2	1
3.16 Filters	—	1	—
3.17 AC generators	—	2	1
3.18 AC motors	—	2	1

MODULE 4. ELECTRONIC FUNDAMENTALS (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 4. ELECTRONICS FUNDAMENTALS	LEVEL		
	A	B1 B3	B2 B2L
4.1 Semiconductors			
4.1.1 Diodes			
(a) Description and characteristics;	—	2	2
(b) Operation and function.	—	—	2
4.1.2 Transistors			
(a) Description and characteristics;	—	1	2
(b) Filters.	—	—	2
4.1.3 Integrated circuits			
(a) Basic description and operation;	—	1	2
(b) Description and operation.	—	—	2
4.2 Printed circuit boards	—	1	2
4.3 Servomechanisms			
(a) Principles;	—	1	2
(b) Construction, operation, and use.	—	—	2

MODULE 5. DIGITAL TECHNIQUES/ELECTRONIC INSTRUMENT SYSTEMS

Regulation (EU) 2023/989

MODULE 5. DIGITAL TECHNIQUES/ELECTRONIC INSTRUMENT SYSTEMS	LEVEL			
	A	B3	B1	B2 B2L
5.1 Electronic instrument systems	1	1	1	1
5.2 Numbering systems	—	—	1	2
5.3 Data conversion	—	—	1	2
5.4 Data buses	—	—	2	2

MODULE 5. DIGITAL TECHNIQUES/ELECTRONIC INSTRUMENT SYSTEMS	LEVEL			
	A	B3	B1	B2 B2L
5.5 Logic circuits				
(a) Identification and applications;	—	—	2	2
(b) Interpretation of logic diagrams.	—	—	—	2
5.6 Basic computer structure				
(a) Computer terminology and technology;	1	1	2	2
(b) Computer operation.	—	—	—	2
5.7 Microprocessors	—	—	—	2
5.8 Integrated circuits	—	—	—	2
5.9 Multiplexing	—	—	—	2
5.10 Fibre optics	—	—	1	2
5.11 Electronic displays	1	1	2	2
5.12 Electrostatic sensitive devices	1	1	2	2
5.13 Software management control	—	1	2	2
5.14 Electromagnetic environment	—	1	2	2
5.15 Typical electronic/digital aircraft systems	1	1	1	1

MODULE 6. MATERIALS AND HARDWARE (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 6. MATERIALS AND HARDWARE	LEVEL		
	A	B1 B3	B2 B2L
6.1 Aircraft materials — ferrous			
(a) Alloy steels used in aircraft;	1	2	1
(b) Testing of ferrous materials;	—	1	1
(c) Repair and inspection procedures.	—	2	1
6.2 Aircraft materials — non-ferrous			
(a) Characteristics;	1	2	1
(b) Testing of non-ferrous materials;	—	1	1
(c) Repair and inspection procedures.	—	2	1
6.3 Aircraft materials — composite and non-metallic			
6.3.1 Composite and non-metallic other than wood and fabric			
(a) Characteristics;	1	2	2
(b) Detection of defects	1	2	—
(c) Repairs and inspection procedures.	—	2	1
6.3.2 Wooden structures	1	1	—
6.3.3 Fabric covering	—	1	—
6.4 Corrosion			
(a) Chemical fundamentals;	1	1	1
(b) Types of corrosion.	2	3	2
6.5 Fasteners			
6.5.1 Screw threads	2	2	2
6.5.2 Bolts, studs and screws	2	2	2
6.5.3 Locking devices	2	2	2
6.5.4 Aircraft rivets	1	2	1

MODULE 6. MATERIALS AND HARDWARE	LEVEL		
	A	B1 B3	B2 B2L
6.6 Pipes and unions			
(a) Identification;	2	2	2
(b) Standard unions.	2	2	1
6.7 Springs	—	2	1
6.8 Bearings	1	2	2
6.9 Transmissions	1	2	2
6.10 Control cables	1	2	1
6.11 Electrical cables and connectors	1	2	2

MODULE 7. MAINTENANCE PRACTICES (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 7. MAINTENANCE PRACTICES	LEVEL		
	A	B1 B3	B2 B2L
7.1 Safety precautions – aircraft and workshop	3	3	3
7.2 Workshop practices	3	3	3
7.3 Tools	3	3	3
7.4 (Reserved)	—	—	—
7.5 Engineering drawings, diagrams and standards	1	2	2
7.6 Fits and Clearances	1	2	1
7.7 Electrical wiring interconnection system (EWIS)	1	3	3
7.8 Riveting	1	2	—
7.9 Pipes and Hoses	1	2	—
7.10 Springs	1	2	—
7.11 Bearings	1	2	—
7.12 Transmissions	1	2	—
7.13 Control cables	1	2	—
7.14 Material handling			
7.14.1 Sheet metal	—	2	—
7.14.2 Composite and non-metallic	—	2	—
7.14.3 Additive manufacturing	1	1	1
7.15 (Reserved)			
7.16 Aircraft weight and balance			
(a) Centre-of-gravity calculation;	—	2	2
(b) Aircraft weighing.	—	2	—
7.17 Aircraft handling and storage	2	2	2
7.18 Disassembly, inspection, repair and assembly techniques			
(a) Types of defects and visual inspection techniques;	2	3	3
(b) General repair methods – structural repair manual;	—	2	—
(c) Non-destructive inspection techniques;	—	2	1
(d) Disassembly and re-assembly techniques;	2	2	2
(e) Troubleshooting techniques.	—	2	2

MODULE 7. MAINTENANCE PRACTICES	LEVEL		
	A	B1 B3	B2 B2L
7.19 Abnormal events			
(a) Inspections following lightning strikes and HIRF penetration;	2	2	2
(b) Inspections following abnormal events such as heavy landings and flight through turbulence.	2	2	—
7.20 Maintenance Procedures	1	2	2
7.21 Documentation & communication	1	2	2

MODULE 8. BASIC AERODYNAMICS (Appendix I to Part-66)

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MODULE 8. BASIC AERODYNAMICS	LEVEL	
	A B3	B1 B2 B2L
8.1 Physics of the atmosphere International Standard Atmosphere (ISA), application to aerodynamics.	1	2
8.2 Aerodynamics	1	2
8.3 Theory of flight	1	2
8.4 High-speed airflow	1	2
8.5 Flight stability and dynamics	1	2

MODULE 9. HUMAN FACTORS (Appendix I to Part-66)

Regulation (EU) 2023/989

MODULE 9. HUMAN FACTORS	LEVEL
	ALL
9.1 General	2
9.2 Human performance and limitations	2
9.3 Social psychology	1
9.4 Factors that affect performance	2
9.5 Physical Environment	1
9.6 Tasks	1
9.7 Communication	2
9.8 Human error	2
9.9 Safety management	2
9.10 The 'Dirty Dozen' and risk mitigation	2

MODULE 10. AVIATION LEGISLATION (Appendix I to Part-66)

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MODULE 10. AVIATION LEGISLATION	LEVEL	
	A	B1 B2 B2L B3
10.1 Regulatory framework	1	1
10.2 Certifying Staff — maintenance	2	2
10.3 Approved maintenance organisations	2	2
10.4 Independent certifying staff	—	3
10.5 Air operations	1	1
10.6 Certification of aircraft, parts, and appliances	2	2
10.7 Continuing airworthiness	2	2
10.8 Oversight principles in continuing airworthiness	1	1
10.9 Maintenance and certification beyond the current EU regulations (if not superseded by EU requirements)	—	1
10.10 Cybersecurity in aviation maintenance	1	1

MODULE 11. TURBINE AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS (Appendix I to Part-66)

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MODULE 11. AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.1 Theory of flight					
(a) Aeroplane aerodynamics and flight controls;	1	1	2	2	1
(b) Aeroplane, other aerodynamic devices.	1	1	2	2	1
11.2 Airframe structures (ATA 51)					
(a) General concepts;	2	2	2	2	2
(b) Airworthiness requirements for structural strength;	2	2	2	2	2
(c) Construction methods.	1	1	2	2	2
11.3 Airframe Structures — aeroplanes					
11.3.1 Fuselage, doors, windows (ATA 52/53/56)					
(a) Construction principles;	1	1	2	2	1
(b) Airborne towing devices;	1	1	1	1	1
(c) Doors.	1	1	2	1	—
11.3.2 Wings (ATA 57)	1	1	2	2	1
11.3.3 Stabilisers (ATA 55)	1	1	2	2	1
11.3.4 Flight control surfaces (ATA 55/57)	1	1	2	2	1
11.3.5 Nacelles/pylons (ATA 54)	1	1	2	2	1
11.4 Air conditioning and cabin pressurisation (ATA 21)					
(a) Pressurisation;	1	1	3	3	—
(b) Air supply;	1	—	3	—	—
(c) Air conditioning;	1	—	3	—	—
(d) Safety and warning devices;	1	1	3	3	—
(e) Heating and ventilation system.	—	1	—	3	1
11.5 Instruments/avionic systems					

MODULE 11. AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.5.1 Instrument systems (ATA 31)	1	1	2	2	2
11.5.2 Avionic Systems Fundamentals of system layouts and operation of: — Autoflight (ATA 22), — Communications (ATA 23), — Navigation Systems (ATA 34).	1	1	1	1	1
11.6 Electrical power (ATA 24)	1	1	3	3	3
11.7 Equipment and furnishings (ATA 25)					
(a) Emergency equipment;	2	2	2	2	2
(b) Cabin and cargo layout.	1	1	1	1	—
11.8 Fire protection (ATA 26)					
(a) Fire and smoke detection system and fire extinguishing systems;	1	1	1	1	—
(b) Portable fire extinguisher.	1	1	1	1	1
11.9 Flight controls (ATA 27)					
(a) Primary and secondary flight controls;	1	1	3	2	2
(b) Actuation and protection;	1	—	3	—	—
(c) System operation;	1	—	3	—	—
(d) Balancing and rigging.	1	1	3	3	2
11.10 Fuel systems (ATA 28, ATA 47)					
(a) Systems layout;	1	1	3	3	1
(b) Fuel handling;	1	1	3	3	1
(c) Indication and warnings;	1	1	3	3	1
(d) Special systems;	1	—	3	—	—
(e) Balancing.	1	—	3	—	—
11.11 Hydraulic power (ATA 29)					
(a) System description;	1	1	3	3	2
(b) System operation (1);	1	1	3	3	2
(c) System operation (2).	1	—	3	—	—
11.12 Ice and rain protection (ATA 30)					
(a) Principles;	1	1	3	3	1
(b) De-icing;	1	1	3	3	1
(c) Anti-icing;	1	—	3	—	—
(d) Wipers;	1	1	3	3	1
(d) Tail protection.	1	—	3	—	—
11.13 Landing Gear (ATA 32)					
(a) Description;	2	2	3	3	2
(b) System operation;	2	2	3	3	2
(c) Air-ground sensing;	2	—	3	—	—
(d) Tail protection.	2	2	3	3	2
11.14 Lights (ATA 33)	2	2	3	3	2
11.15 Oxygen (ATA 35)	1	1	3	3	2
11.16 Pneumatic/vacuum (ATA 36)					
(a) Systems;	1	1	3	3	2
(b) Pumps.	1	1	3	3	2
11.17 Water/waste (ATA 38)					

MODULE 11. AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
(a) Systems;	2	2	3	3	2
(b) Corrosion.	2	2	3	3	2
11.18 On-board maintenance systems (ATA 45)	1	—	2	—	—
11.19 Integrated Modular Avionics (ATA 42)					
(a) Overall system description and theory;	1	—	2	—	—
(b) Typical system layouts.	1	—	2	—	—
11.20 Cabin systems (ATA 44)	1	—	2	—	—
11.21 Information systems (ATA 46)	1	—	2	—	—

MODULE 12. HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS (Appendix I to Part-66)

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MODULE 12. HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
12.1 Theory of flight — rotary wing aerodynamics	1	2
12.2 Flight control systems (ATA 67)	2	3
12.3 Blade tracking and vibration analysis (ATA 18)	1	3
12.4 Transmission	1	3
12.5 Airframe structures (ATA 51)		
(a) General concept;	2	2
(b) Construction methods of the principal elements.	1	2
12.6 Air conditioning (ATA 21)		
12.6.1 Air supply	1	2
12.6.2 Air conditioning	1	3
12.7 Instruments/avionic systems		
12.7.1 Instrument systems (ATA 31)	1	2
12.7.2 Avionic systems	1	1
Fundamentals of system layouts and operation of: Auto Flight (ATA 22); Communications (ATA 23); Navigation Systems (ATA 34).		
12.8 Electrical Power (ATA 24)	1	3
12.9 Equipment and furnishings (ATA 25)		
(a) Emergency equipment; Seats, harnesses, and belts; Lifting systems;	2	2
(b) Emergency flotation systems; Cabin layout, cargo retention; Equipment layout; Cabin furnishing installation.	1	1
12.10 Fire Protection (ATA 26)		
(a) Fire and smoke detection systems and Fire-extinguishing systems;	1	3
(b) Portable fire extinguishers.	1	1
12.11 Fuel Systems (ATA 28)	1	3

MODULE 12. HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
12.12 Hydraulic Power (ATA 29)	1	3
12.13 Ice and Rain Protection (ATA 30)	1	3
12.14 Landing Gear (ATA 32)		
(a) System description and operation;	2	3
(b) Sensors.	2	3
12.15 Lights (ATA 33)	2	3
12.16 (Reserved)	2	3
12.17 Integrated Modular Avionics (ATA 42)		
(a) Overall system description and theory	1	2
(b) Typical system layouts	1	2
12.18 On-board Maintenance Systems (ATA 45)	1	2
Central maintenance computers; Data-loading system; Electronic library system.		
12.19 Information Systems (ATA 46)	1	2

MODULE 13. AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS (Appendix I to Part-66)

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C/N: Communication & Navigation; Ins.: Instruments; A/F: Autoflight; Sur.: Surveillance; A/S: Airframe & Systems

MODULE 13. AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
13.1 Theory of Flight							
(a) Aeroplane Aerodynamics and Flight Controls;	1	1	—	—	—	—	—
(b) Rotary Wing Aerodynamics.	1	1	—	—	—	—	—
13.2 Structures – General Concepts (ATA 51)							
(a) General concept;	2	2	—	—	—	—	—
(b) Fundamentals of structural systems;	1	1	—	—	—	—	—
13.3 Autoflight (ATA 22)							
(a) Fundamentals of automatic flight control;	3	—	—	—	3	—	—
(b) Autothrottle systems and automatic landing systems.	3	—	—	—	3	—	—
13.4 Communication/Navigation (ATA 23/34)							
(a) Fundamentals of communication and navigation systems;	3	—	3	—	—	—	—
(b) Fundamentals of aircraft surveillance systems.	3	—	—	—	—	3	—
13.5 Electrical power (ATA 24)	3	3	—	—	—	—	—
13.6 Equipment and furnishings (ATA 25)	3	—	—	—	—	—	—

MODULE 13. AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
13.7 Flight Controls							
(a) Primary and secondary flight controls (ATA 27);	2	—	—	—	2	—	—
(b) Actuation and protection;	2	—	—	—	2	—	—
(c) System operation;	3	—	—	—	3	—	—
(d) Rotorcraft flight controls (ATA 67).	2	—	—	—	2	—	—
13.8 Instruments (ATA 31)	3	—	—	3	—	—	—
13.9 Lights (ATA 33)	3	3	—	—	—	—	—
13.10 On-board maintenance systems (ATA 45)	3	—	—	—	—	—	—
13.11 Air conditioning and cabin pressurisation (ATA 21)							
(a) Pressurisation;	3	—	—	—	—	—	3
(b) Air supply;	1	—	—	—	—	—	1
(c) Air conditioning;	3	—	—	—	—	—	3
(d) Safety and warning devices.	3	—	—	—	—	—	3
13.12 Fire protection (ATA 26)							
(a) Fire and smoke detection system and fire-extinguishing systems;	3	—	—	—	—	—	3
(b) Portable fire extinguisher.	1	—	—	—	—	—	1
13.13 Fuel systems (ATA 28, ATA 47)							
(a) System layout;	1	—	—	—	—	—	1
(b) Fuel handling;	2	—	—	—	—	—	2
(c) Indications and warnings;	3	—	—	—	—	—	3
(d) Special systems;	1	—	—	—	—	—	1
(e) Balancing	3	—	—	—	—	—	3
13.14 Hydraulic power (ATA 29)							
(a) System layout;	1	—	—	—	—	—	1
(b) System operation (1);	3	—	—	—	—	—	3
(c) System operation (2);	3	—	—	—	—	—	3
13.15 Ice and rain protection (ATA 30)							
(a) Principles;	2	—	—	—	—	—	2
(b) De-icing;	3	—	—	—	—	—	3
(c) Anti-icing;	2	—	—	—	—	—	2
(d) Wiper systems;	1	—	—	—	—	—	1
(e) Rain repellent	1	—	—	—	—	—	1
13.16 Landing gear (ATA 32)							
(a) Description;	1	—	—	—	—	—	1
(b) System;	3	—	—	—	—	—	3
(c) Air-ground sensing.	3	—	—	—	—	—	3
13.17 Oxygen (ATA 35)	3	—	—	—	—	—	3
13.18 Pneumatic/vacuum (ATA 36)	2	—	—	—	—	—	2
13.19 Water/waste (ATA 38)	2	—	—	—	—	—	2
13.20 Integrated modular avionics (ATA 42)							
(a) Overall system description and theory;	3	-	—	—	—	—	—

MODULE 13. AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
(b) Typical system layouts.	3	-	—	—	—	—	—
13.21 Cabin systems (ATA 44)	3	—	—	—	—	—	—
13.22 Information systems (ATA 46)	3	—	—	—	—	—	—

MODULE 14. PROPULSION (Appendix I to Part-66)

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MODULE 14. PROPULSION	LEVEL
	B2 B2L Instruments B2L Airframe & Systems
14.1 Engines	
(a) Turbine engines;	1
(b) Auxiliary power units (APUs);	1
(c) Piston engines;	1
(d) Electric and hybrid engines;	2
(e) Engine control.	2
14.2 Electric/electronic engine indication systems	2
14.3 Propeller systems	2
14.4 Starting and ignition systems	2

MODULE 15. GAS TURBINE ENGINE (Appendix I to Part-66)

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MODULE 15. GAS TURBINE ENGINE	LEVEL	
	A1 A3	B1.1 B1.3
15.1 Fundamentals	1	2
15.2 Engine performance	—	2
15.3 Inlet	2	2
15.4 Compressors	1	2
15.5 Combustion section	1	2
15.6 Turbine section	2	2
15.7 Exhaust	1	2
15.8 Bearings and seals	—	2
15.9 Lubricants and fuels	1	2
15.10 Lubrication systems	1	2
15.11 Fuel systems	1	2
15.12 Air systems	1	2
15.13 Starting and ignition systems	1	2
15.14 Engine indication systems	1	2
15.15 Alternate turbine constructions	—	1
15.16 Turboprop engines	1	2
15.17 Turboshift engines	1	2
15.18 Auxiliary power units (APUs)	1	2

MODULE 15. GAS TURBINE ENGINE	LEVEL	
	A1 A3	B1.1 B1.3
15.19 Power plant installation	1	2
15.20 Fire protection systems	1	2
15.21 Engine monitoring and ground operation	1	3
15.22 Engine storage and preservation	—	2

MODULE 16. PISTON ENGINE (Appendix I to Part-66)

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MODULE 16. PISTON ENGINE	LEVEL	
	A2 A4	B1.2 B1.4 B3
16.1 Fundamentals	1	2
16.2 Engine performance	1	2
16.3 Engine construction	1	2
16.4 Engine fuel systems		
16.4.1 Carburettors	1	2
16.4.2 Fuel injection systems	1	2
16.4.3 Electronic engine control	1	2
16.5 Starting and ignition systems	1	2
16.6 Induction, exhaust and cooling systems	1	2
16.7 Supercharging/turbocharging	1	2
16.8 Lubricants and fuels	1	2
16.9 Lubrication systems	1	2
16.10 Engine indication systems	1	2
16.11 Power plant installation	1	2
16.12 Engine monitoring and ground operation	1	3
16.13 Engine storage and preservation	—	2
16.14 Alternative piston engine constructions	1	1

MODULE 17. PROPELLER (Appendix I to Part-66)

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MODULE 17A. PROPELLER	LEVEL	
	A1 A2	B1.1 B1.2 B3
17.1 Fundamentals	1	2
17.2 Propeller construction	1	2
17.3 Propeller pitch control	1	2
17.4 Propeller synchronising	—	2
17.5 Propeller ice protection	1	2
17.6 Propeller maintenance	1	3
17.7 Propeller storage and preservation	1	2'

3. Basic training methods

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An appropriate training method, or combination of methods, shall be determined for the entire course or for each of its modules or submodules, with regard to the scope and objectives of each training phase and taking into consideration the benefits and limitations of the available training methods.

Multimedia-based training (MBT) methods may be used in order to achieve the training objectives either in a physically or in a virtually controlled environment.

AMC1 Appendix I — Basic Knowledge Requirements (except for category L licence) Section 2

ED Decision 2023/019/R

MODULARISATION

MODULE 1 — MATHEMATICS

MODULE 1 — MATHEMATICS	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	B3
1.1 <i>Arithmetic</i> Arithmetical terms and signs, methods of multiplication and division, fractions and decimals, factors and multiples, weights, measures and conversion factors, ratio and proportion, averages and percentages, areas and volumes, squares, cubes, square and cube roots.	1	2
1.2 <i>Algebra</i> (a) Evaluating simple algebraic expressions, addition, subtraction, multiplication and division, use of brackets, simple algebraic fractions;	1	2
(b) Linear equations and their solutions; Indices and powers, negative and fractional indices; Binary and other applicable numbering systems; Simultaneous equations and second-degree equations with one unknown; Logarithms.	—	1
1.3 <i>Geometry</i> (a) Simple geometrical constructions;	—	1
(b) Graphical representation: nature and uses of graphs, graphs of equations/functions;	2	2
(c) Simple trigonometry: trigonometrical relationships, use of tables and rectangular and polar coordinates.	—	2

MODULE 2 — PHYSICS

MODULE 2 — PHYSICS	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	
	B3	
2.1 Matter Nature of matter: the chemical elements, structure of atoms, molecules; Chemical compounds; States: solid, liquid, and gaseous; Changes between states.	1	2
2.2 Mechanics 2.2.1 Statics Forces, moments and couples, representation as vectors; Centre of gravity; Elements of theory of stress, strain, and elasticity: tension, compression, shear, and torsion; Nature and properties of solid, fluid, and gas matter; Pressure and buoyancy in liquids (barometers).	1	2
2.2.2 Kinetics Linear movement: uniform motion in a straight line, motion under constant acceleration (motion under gravity); Rotational movement: uniform circular motion (centrifugal/centripetal forces); Periodic motion: pendular movement; Simple theory of vibration, harmonics, and resonance; Velocity ratio, mechanical advantage, and efficiency.	1	2
2.2.3 Dynamics (a) Mass; Force, inertia, work, power, energy (potential, kinetic and total energy), heat, efficiency; (b) Momentum, conservation of momentum; Impulse; Gyroscopic principles; Friction: nature and effects, coefficient of friction (rolling resistance).	1	2
	1	2
2.2.4 Fluid dynamics (a) Specific gravity and density; (b) Viscosity, fluid resistance, effects of streamlining; Effects of compressibility on fluids; Static, dynamic, and total pressure: Bernoulli's Theorem, venturi.	2	2
	1	2
2.3 Thermodynamics (a) Temperature: thermometers and temperature scales (Celsius, Fahrenheit and Kelvin); definition of heat; (b) Heat capacity, specific heat;	2	2
	1	2

MODULE 2 — PHYSICS	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	
	B3	
Heat transfer: convection, radiation and conduction; Volumetric expansion; First and second law of thermodynamics; Gases: ideal gases laws; specific heat at constant volume and constant pressure, work done by expanding gas; Isothermal, adiabatic expansion and compression, engine cycles, constant volume and constant pressure, refrigerators and heat pumps; Latent heats of fusion and evaporation, thermal energy, heat of combustion.		
2.4 Optics (Light) Nature of light; speed of light; Laws of reflection and refraction: reflection at plane surfaces, reflection by spherical mirrors, refraction, lenses; Fibre optics.	—	2
2.5 Wave motion and sound Wave motion: mechanical waves, sinusoidal wave motion, interference phenomena, standing waves; Sound: speed of sound, production of sound, intensity, pitch and quality, Doppler effect.	—	2

MODULE 3 — ELECTRICS FUNDAMENTALS

MODULE 3 — ELECTRICS FUNDAMENTALS	LEVEL		
	A1 A2 A3 A4	B1 B2 B2L	B3
3.1 Electron theory Structure and distribution of electrical charges within atoms, molecules, ions, and compounds; Molecular structure of conductors, semiconductors, and insulators.	1	1	1
3.2 Static electricity and conduction Static electricity and distribution of electrostatic charges; Electrostatic laws of attraction and repulsion; Units of charge, Coulomb's law; Conduction of electricity in solids, liquids, gases and in vacuum.	1	2	1
3.3 Electrical terminology The following terms, their units and factors affecting them: potential difference, electromotive force, voltage, current, resistance, conductance, charge, conventional current flow, electron flow.	1	2	1
3.4 Generation of electricity Production of electricity by the following methods: light, heat, friction, pressure, chemical reaction, magnetism, and motion.	1	1	1
3.5 Sources of DC electricity Construction and basic chemical reaction of primary cells, secondary cells, lead acid cells, nickel cadmium cells, lithium cells, nickel cells and other alkaline cells; Cells connected in series and in parallel; Internal resistance and its effect on a battery; Construction, materials, and operation of thermocouples; Operation of photocells.	1	2	2
3.6 DC circuits Ohm's law, Kirchhoff's voltage, and current laws; Calculations using the above laws to find resistance, voltage, and current; Significance of the internal resistance of a supply.	1	2	1
3.7 Resistance/Resistor (a) Resistance Specific resistance; Calculation of total resistance using series, parallel and series-parallel combinations; Operation and use of potentiometers and rheostats; Operation of Wheatstone Bridge.	—	2	1
(b) Resistor Positive and negative temperature coefficient conductance; Resistor colour code, values and tolerances, preferred values, wattage ratings; Resistors in series and in parallel; Fixed resistors, stability, tolerance and limitations, methods of construction; Variable resistors, thermistors, voltage-dependent resistors; Construction of potentiometers and rheostats; Construction of Wheatstone Bridge.	—	1	—

MODULE 3 — ELECTRICS FUNDAMENTALS	LEVEL		
	A1 A2 A3 A4	B1 B2 B2L	B3
3.8 Power Power, work, and energy (kinetic and potential); Dissipation of power by a resistor; Power formula; Calculations involving power, work, and energy.	—	2	1
3.9 Capacitance/Capacitor Operation and function of a capacitor; Factors that affect the capacitance area of plates, distance between plates, number of plates, dielectric and dielectric constant, working voltage, voltage rating; Capacitor types, construction, and function; Capacitor colour-coding; Calculations of capacitance and voltage in series and in parallel circuits; Exponential charge and discharge of a capacitor, time constants; Testing of capacitors.	—	2	1
3.10 Magnetism (a) Theory of magnetism; Properties of a magnet; Action of a magnet suspended in the Earth's magnetic field; Magnetisation and demagnetisation; Magnetic shielding; Various types of magnetic material; Electromagnet construction and principles of operation; Handclasp rules to determine magnetic field around current-carrying conductor.	—	2	1
	—	2	1
(b) Magnetomotive force, field strength, magnetic flux density, permeability, hysteresis loop, retentivity, coercive force reluctance, saturation point, eddy currents; Precautions for care and storage of magnets.	—	2	1
3.11 Inductance/Inductor Faraday's law; Action of inducing a voltage in a conductor that moves in a magnetic field; Induction principles; Effects of the following on the magnitude of an induced voltage: magnetic field strength, rate of change of flux, number of conductor turns; Mutual induction; The effect that the rates of change of primary current and mutual inductance have on induced voltage; Factors that affect mutual inductance: number of turns in the coil, physical size of the coil, permeability of the coil, position of coils with respect to each other; Lenz's law and polarity determining rules; Back EMF, self-induction; Saturation point; Principal uses of inductors.	—	2	1

MODULE 3 — ELECTRICS FUNDAMENTALS	LEVEL		
	A1 A2 A3 A4	B1 B2 B2L	B3
3.12 DC motor/generator theory Basic motor and generator theory; Construction and purpose of components in a DC generator; Operation of and factors that affect the output and direction of the current in DC generators; Operation of and factors that affect the output power, torque, speed, and direction of rotation of DC motors; Series-wound, shunt-wound and compound motors; Starter generator construction.	—	2	1
3.13 AC theory Sinusoidal waveform: phase, period, frequency, cycle; Instantaneous, average, root mean square, peak, peak-to-peak current values and calculations of these values in relation to voltage, current and power; Triangular/Square waves; Single-phase/Three-phase principles.	1	2	1
3.14 Resistive (R), capacitive (C) and inductive (L) circuits Phase the relationship of voltage and current in L, C and R circuits, parallel, series and series-parallel; Power dissipation in L, C and R circuits; Impedance, phase angle, power factor and current calculations; True power, apparent power, and reactive power calculations.	—	2	1
3.15 Transformers Transformer construction principles and operation; Transformer losses and methods for overcoming them; Transformer action under load and no-load conditions; Power transfer, efficiency, polarity markings; Line and phase voltages and currents; Power in a three-phase system; Primary and secondary current, voltage, turn ratio, power, efficiency; Auto-transformers.	—	2	1
3.16 Filters Operation, application, and uses of the following filters: low pass, high pass, band pass, band stop.	—	1	—
3.17 AC generators Rotation of loop in a magnetic field and waveform produced; Operation and construction of revolving armature and revolving field type AC generators; Single-phase, two-phase, and three-phase alternators; Three-phase star and delta connection advantages, and uses; Permanent magnet generators.	—	2	1
3.18 AC motors Construction, principles of operation and characteristics of: AC synchronous and induction motors both single-phase and polyphase; Methods of speed control and direction of rotation; Methods of producing a rotating field: capacitor, shaded or split pole.	—	2	1

MODULE 4 — ELECTRONICS FUNDAMENTALS

MODULE 4 — ELECTRONICS FUNDAMENTALS	LEVEL		
	A1		
	A2	B1	B2
	A3	B3	B2L
	A4		
4.1 Semiconductors			
4.1.1 Diodes			
(a) Description and characteristics	—	2	2
Diode symbols;			
Diode characteristics and properties;			
Diodes in series and in parallel;			
Materials, electron configuration, electrical properties;			
P and N type materials: effects of impurities on conduction, majority and minority characters;			
P–N junction in a semiconductor, development of a potential across a P–N junction in unbiased, forward-biased and reverse-biased conditions;			
Diode parameters: peak inverse voltage, maximum forward current, temperature, frequency, leakage current, power dissipation;			
Main characteristics and use of silicon-controlled rectifiers (thyristors), light-emitting diodes (LEDs), photo-conductive diodes, rectifier diodes.			
(b) Operation and function	—	—	2
Operation and function of diodes in the following circuits: clippers, clampers, full- and half-wave rectifiers, bridge rectifiers, voltage doublers and triplers;			
Detailed operation and characteristics of the following devices: silicon-controlled rectifier (thyristor), light-emitting diode (LED), Schottky diode, photo-conductive diode, varactor diode, varistor, rectifier diodes, Zener diode.			
Functional testing of diodes.			
4.1.2 Transistors			
(a) Description and characteristics	—	1	2
Transistor symbols;			
Component description and orientation;			
Transistor characteristics and properties.			
(b) Construction and operation	—	—	2
Construction and operation of PNP and NPN transistors;			
Base, collector and emitter configurations;			
Testing of transistors;			
Basic appreciation of other transistor types, including types of FET and their uses;			
Application of transistors: amplifier classes (A, B, C);			

MODULE 4 — ELECTRONICS FUNDAMENTALS	LEVEL		
	A1	B1	B2
	A2	B3	B2L
	A3		
	A4		
Simple circuits including bias, decoupling, feedback and stabilisation; Multistage circuit principles: cascades, push–pull, oscillators, multivibrators, flip-flop circuits; Operation and amplifier stages connecting methods: resistive, capacitive, direct, inverting, non-inverting and adding.			
4.1.3 Integrated circuits			
(a) Description and operation of logic circuits and linear circuits/operational amplifiers;	—	1	2
(b) Introduction to the operation and function of an operational amplifier used as: an integrator, a differentiator, a voltage follower, a comparator; Advantages and disadvantages of positive and negative feedback.	—	—	2
4.2 Printed circuit boards Description and use of printed circuit boards.	—	1	2
4.3 Servomechanisms			
(a) Principles Understanding of the following principles: open- and closed-loop systems, servomechanism, feedback, follow-up, null, overshoot, damping, deadband, hunting, proximity switches, analogue transducers, synchro systems and components, digital tachometers and encoders, inductance, and capacitance transmitters;	—	1	2
(b) Construction operation and use of the following synchro-system components: resolvers, differential, control and torque, E and I transformers, inductance transmitters, capacitance transmitters, synchronous transmitters; Construction, operation and use of servomechanism and PID controller; Fault-finding of servo defects, reversal of synchro leads, hunting.	—	—	2

MODULE 5 — DIGITAL TECHNIQUES / ELECTRONIC INSTRUMENT SYSTEMS

MODULE 5 — DIGITAL TECHNIQUES / ELECTRONIC INSTRUMENT SYSTEMS	LEVEL			
	A1	B3	B1	B2 B2L
	A2			
	A3			
	A4			
5.1 <i>Electronic instrument systems</i> Typical arrangements of systems and cockpit layout of electronic instrument systems.	1	1	1	1
5.2 <i>Numbering systems</i> Numbering systems: binary, octal, and hexadecimal; Demonstration of conversions between the decimal and binary systems, octal and hexadecimal systems and vice versa.	—	—	1	2
5.3 <i>Data conversion</i> Analogue data, Digital data; Operation and application of analogue-to-digital and digital-to-analogue converters, inputs and outputs, limitations of various types.	—	—	1	2
5.4 <i>Data buses</i> Operation of data buses in aircraft systems, including knowledge of ARINC and other specifications. Aircraft network/Ethernet.	—	—	2	2
5.5 <i>Logic circuits</i> (a) Identification of common logic gate symbols, tables and equivalent circuits; Applications used for aircraft systems, schematic diagrams.	—	—	2	2
(b) Interpretation of logic diagrams.	—	—	—	2
5.6 <i>Basic computer structure</i> (a) Computer terminology (including bit, byte, software, hardware, CPU, IC, and various memory devices such as RAM, ROM, PROM); Computer technology (as applied in aircraft systems).	1	1	2	2
(b) Computer operation, layout, and interface of the major components in a microcomputer, including their associated bus systems; Information contained in single- and multi-address instruction words; Memory-associated terms; Operation of typical memory devices; Operation, advantages, and disadvantages of the various data storage systems.	—	—	—	2
5.7 <i>Microprocessors</i> Functions performed and overall operation of a microprocessor; Basic operation of each of the following microprocessor elements: control and processing unit, clock, register, arithmetic logic unit.	—	—	—	2
5.8 <i>Integrated circuits</i> Operation and use of encoders and decoders; Function of encoder types.	—	—	—	2

MODULE 5 — DIGITAL TECHNIQUES / ELECTRONIC INSTRUMENT SYSTEMS	LEVEL			
	A1 A2 A3 A4	B3	B1	B2 B2L
5.9 <i>Multiplexing</i> Operation, application and identification in logic diagrams of multiplexers and demultiplexers.	—	—	—	2
5.10 <i>Fibre optics</i> Advantages and disadvantages of fibre optic data transmission over electrical wire propagation; Fibre optic data bus; Fibre-optic-related terms; Terminations; Couplers, control terminals, remote terminals; Application of fibre optics in aircraft systems.	—	—	1	2
5.11 <i>Electronic displays</i> Principles of operation of common types of displays used in modern aircraft, including cathode-ray tubes (CRTs), light-emitting diodes (LEDs) and liquid crystal displays (LCDs).	1	1	2	2
5.12 <i>Electrostatic-sensitive devices</i> Special handling of components sensitive to electrostatic discharges; Awareness of risks and possible damage, component, and personnel antistatic protection devices.	1	1	2	2
5.13 <i>Software management control</i> Awareness of restrictions, airworthiness requirements and possible catastrophic effects of unapproved changes to software programs.	—	1	2	2
5.14 <i>Electromagnetic environment</i> Influence of the following phenomena on maintenance practices for electronic systems: EMC — Electromagnetic Compatibility, EMI — Electromagnetic Interference, HIRF — High-Intensity Radiated Field, Lightning / lightning protection.	—	1	2	2
5.15 <i>Typical electronic/digital aircraft systems</i> General arrangement of typical electronic/digital aircraft systems and associated BITE (Built-In Test Equipment), such as: (a) ACARS — ARINC Communication and Addressing and Reporting System, FBW — Fly-by-Wire, FMS — flight management system, IRS — inertial reference system; (b) ECAM — electronic centralised aircraft monitoring, EICAS — engine indication and crew alerting system, EFIS — electronic flight instrument system, GNSS — global navigation satellite system, TCAS — traffic alert collision avoidance system, Integrated Modular Avionics, Cabin Systems, Information Systems.	1	1	1	1

MODULE 6 — MATERIALS AND HARDWARE

MODULE 6 — MATERIALS AND HARDWARE	LEVEL		
	A1		
	A2	B1	B2
	A3	B3	B2L
	A4		
6.1 Aircraft materials — ferrous			
(a) Characteristics, properties and identification of common alloy steels used in aircraft; Heat treatment and application of alloy steels.	1	2	1
(b) Testing of ferrous materials for hardness, tensile strength, fatigue strength and impact resistance.	—	1	1
(c) Repair and inspection procedures for ferrous materials, structures, and airframes.	—	2	1
6.2 Aircraft materials — non-ferrous			
(a) Characteristics, properties and identification of common non-ferrous materials used in aircraft; Heat treatment and application of non-ferrous materials.	1	2	1
(b) Testing of non-ferrous material for hardness, tensile strength, fatigue strength and impact resistance.	—	1	1
(c) Repair and inspection procedures for non-ferrous materials, structures, and airframes.	—	2	1
6.3 Aircraft materials — composite and non-metallic			
6.3.1 Composite and non-metallic materials other than wood and fabric			
(a) Characteristics, properties and identification of common composite and non-metallic materials, other than wood, used in aircraft; Sealant and bonding agents.	1	2	2
(b) Detection of defects/deterioration in composite and non-metallic materials.	1	2	—
(c) Repair of and inspection procedures for composite and non-metallic materials, structures, and airframes.	—	2	1
6.3.2 Wooden structures	1	1	—
Construction methods of wooden airframe structures; Characteristics, properties and types of wood and glue used in aeroplanes; Preservation and maintenance of wooden structures; Types of defects in wood material and wooden structures; Detection of defects in wooden structures; Repair of wooden structures.			
6.3.3 Fabric covering	—	1	—
Characteristics, properties and types of fabrics used in aeroplanes; Inspection methods for fabrics; Types of defects in fabrics; Repair of fabric covering.			

MODULE 6 — MATERIALS AND HARDWARE	LEVEL		
	A1	B1	B2
	A2		
	A3		
	A4		
6.4 <i>Corrosion</i>			
(a) Chemical fundamentals; Formation by galvanic action process, microbiological contamination, mechanical stress.	1	1	1
(b) Types of corrosion and their identification; Causes of corrosion; Material types, and their susceptibility to corrosion.	2	3	2
6.5 <i>Fasteners</i>			
6.5.1 <i>Screw threads</i> Screw nomenclature; Thread forms, dimensions and tolerances for standard threads used in aircraft; Measuring screw threads.	2	2	2
6.5.2 <i>Bolts, studs, and screws</i> Bolt types: specification, identification and marking of aircraft bolts, international standards; Nuts: self-locking, anchor, standard types; Machine screws: aircraft specifications; Studs: types and uses, insertion, and removal; Self-tapping screws, dowels.	2	2	2
6.5.3 <i>Locking devices</i> Tab and spring washers, locking plates, split pins, pal-nuts, wire locking, quick-release fasteners, keys, circlips, cotter pins.	2	2	2
6.5.4 <i>Aircraft rivets</i> Types of solid and blind rivets: specifications and identification, heat treatment.	1	2	1
6.6 <i>Pipes and unions</i>			
(a) Identification and types of rigid and flexible pipes and their connectors used in aircraft;	2	2	2
(b) Standard unions for aircraft hydraulic, fuel, oil, pneumatic and air system pipes.	2	2	1
6.7 <i>Springs</i> Types of springs, materials, characteristics, and applications.	—	2	1
6.8 <i>Bearings</i> Purpose of bearings, loads, material, construction; Types of bearings and their application.	1	2	2
6.9 <i>Transmissions</i> Gear types and their application; Gear ratios, reduction and multiplication gear systems, driven and driving gears, idler gears, mesh patterns; Belts and pulleys, chains and sprockets.	1	2	2

MODULE 6 — MATERIALS AND HARDWARE	LEVEL		
	A1		
	A2	B1	B2
	A3	B3	B2L
	A4		
6.10 <i>Control cables</i> Types of cables; End fittings, turnbuckles and compensation devices; Pulleys and cable system components; Bowden cables; Aircraft flexible control systems.	1	2	1
6.11 <i>Electrical cables and connectors</i> Cable types, construction and characteristics; High-tension and coaxial cables; Crimping; Connector types, pins, plugs, sockets, insulators, current and voltage rating, coupling, identification codes.	1	2	2

MODULE 7 — MAINTENANCE PRACTICES

MODULE 7 — MAINTENANCE PRACTICES	LEVEL		
	A1	B1	B2
	A2	B3	B2L
	A3		
	A4		
7.1 Safety precautions — aircraft and workshop Aspects of safe working practices including precautions to be taken when working with electricity, gases (especially oxygen), oils, and chemicals. Fuel tank safety and fuel tank entry procedures and precautions. Awareness and precautions regarding aircraft equipped with ballistic recovery systems. Also, instructions for the remedial action to be taken in the event of a fire or another accident with one or more of these hazards, including information on fire-extinguishing agents.	3	3	3
7.2 Workshop practices Care of tools, control of tools, use of workshop materials; Dimensions, allowances and tolerances, workmanship standards; Calibration of tools and equipment, calibration standards.	3	3	3
7.3 Tools Common hand-tool types; Common power-tool types; Operation and use of precision-measuring tools; Lubrication equipment and methods; Operation, function, and use of electrical general test equipment.	3	3	3
7.4 (Reserved)			
7.5 Engineering drawings, diagrams, and standards Drawing types and diagrams, their symbols, dimensions, tolerances and projections; Identification of title block information; Microfilm, microfiche, and computerised presentations; Specification 100 of the Air Transport Association (ATA) of America; Aeronautical and other applicable standards including ISO, AN, MS, NAS and MIL; Wiring diagrams and schematic diagrams.	1	2	2
7.6 Fits and clearances Drill sizes for bolt holes, classes of fits; Common system for fits and clearances; Schedule of fits and clearances for aircraft and engines; Limits for bow, twist and wear; Standard methods for checking shafts, bearings, and other parts.	1	2	1
7.7 Electrical wiring interconnection system (EWIS) Continuity, insulation and bonding techniques and testing; Use of crimp tools: hand and hydraulic operated; Testing of crimp joints; Connector pin removal and insertion; Coaxial cables: testing and installation precautions; Identification of wire types, their inspection criteria and damage tolerance; Wiring protection techniques: cable looming and loom support, cable clamps, protective sleeving techniques including heat shrink wrapping, shielding; High-Intensity Radiated Fields (HIRF) and protection principles; Soldering of electrical wires, EWIS installations, inspection, repair, maintenance, and cleanliness standards.	1	3	3

MODULE 7 — MAINTENANCE PRACTICES	LEVEL		
	A1	B1	B2
	A2	B3	B2L
	A3		
	A4		
7.8 Riveting Riveted joints, rivet spacing and pitch; Tools used for riveting and dimpling; Inspection of riveted joints.	1	2	—
7.9 Pipes and hoses Bending and belling/flaring aircraft pipes; Inspection and testing of aircraft pipes and hoses; Installation and clamping of pipes.	1	2	—
7.10 Springs Inspection and testing of springs.	1	2	—
7.11 Bearings Testing, cleaning and inspection of bearings; Lubrication requirements for bearings; Defects in bearings and their causes.	1	2	—
7.12 Transmissions Inspection of gears, backlash; Inspection of belts and pulleys, chains and sprockets; Inspection of screw jacks, lever devices, push–pull rod systems.	1	2	—
7.13 Control cables Swaging of end fittings; Inspection and testing of control cables; Bowden cables; aircraft flexible control systems.	1	2	—
7.14 Material handling			
7.14.1 Sheet metal Marking out and calculation of bend allowance; Sheet metal working, including bending and forming; Inspection of sheet metal work.	—	2	—
7.14.2 Composite and non-metallic Bonding practices; Environmental conditions; Inspection methods.	—	2	—
7.14.3 Additive manufacturing Common additive manufacturing techniques and their influence on the mechanical properties of the finished part; Inspection of additive manufactured parts and common production failures.	1	1	1
7.15 (Reserved)			
7.16 Aircraft weight and balance			
(a) Calculation of centre-of-gravity / balance limits: use of relevant documents.	—	2	2
(b) Preparation of aircraft for weighing; Aircraft weighing.	—	2	—
7.17 Aircraft handling and storage Aircraft taxiing/towing and associated safety precautions; Aircraft jacking, chocking, securing and associated safety precautions; Aircraft storage methods;	2	2	2

MODULE 7 — MAINTENANCE PRACTICES	LEVEL		
	A1	B1	B2
	A2	B3	B2L
	A3		
	A4		
Refuelling/defuelling procedures; De-icing/anti-icing procedures; Electrical, hydraulic, and pneumatic ground supplies; Effects of environmental conditions on aircraft handling and operation.			
7.18 Disassembly, inspection, repair, and assembly techniques			
(a) Types of defects and visual inspection techniques; Corrosion removal, assessment and reprotection;	2	3	3
(b) General repair methods, structural repair manual; Ageing, fatigue, and corrosion control programmes;	–	2	–
(c) Non-destructive inspection techniques including penetrant, radiographic, eddy current, magnetic particle, ultrasonic and borescope inspections; including practical training in colour contrast penetrant inspection;	–	2	1
(d) Disassembly and reassembly techniques;	2	2	2
(e) Troubleshooting techniques.	–	2	2
7.19 Abnormal events			
(a) Inspections following lightning strikes and HIRF penetration;	2	2	2
(b) Inspections following abnormal events such as heavy landings and flight through turbulence.	2	2	–
7.20 Maintenance procedures	1	2	2
Maintenance planning; Modification procedures; Stores procedures; Certification/release procedures; Interface with aircraft operation; Maintenance Inspection / Quality Control / Quality Assurance; Additional maintenance procedures; Control of life-limited components.			
7.21 Documentation and communication	1	2	2
Documentation: elements and criteria for writing work reports, troubleshooting reports, and shift handover instructions. Communication: clear, comprehensive, and concise.			

MODULE 8 — BASIC AERODYNAMICS

MODULE 8 — BASIC AERODYNAMICS	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	
	B3	
8.1 <i>Physics of the atmosphere</i> International Standard Atmosphere (ISA), and its application to aerodynamics.	1	2
8.2 <i>Aerodynamics</i> Airflow around a body; Boundary layer, laminar and turbulent flow, free stream flow, relative airflow, upwash and downwash, vortices, stagnation; The terms: camber, chord, mean aerodynamic chord, profile (parasite) drag, induced drag, centre of pressure, angle of attack, wash-in and wash-out, fineness ratio, wing shape and aspect ratio; Thrust, weight, aerodynamic resultant; Generation of lift and drag angle of attack, lift coefficient, drag coefficient, polar curve, stall; Aerofoil contamination including ice, snow, and frost.	1	2
8.3 <i>Theory of flight</i> Relationship between lift, weight, thrust and drag; Glide ratio; Steady-state flights, performance; Theory of the turn; Influence of load factor: stall, flight envelope, and structural limitations; Lift augmentation.	1	2
8.4 <i>High-speed airflow</i> Speed of sound, subsonic flight, transonic flight, supersonic flight, Mach number, critical Mach number, compressibility buffet, shock wave, aerodynamic heating, area rule; Factors that affect airflow in engine intakes of high-speed aircraft; Effects of sweepback on critical Mach number.	1	2
8.5 <i>Flight stability and dynamics</i> Longitudinal, lateral, and directional stability (active and passive).	1	2

MODULE 9 — HUMAN FACTORS

MODULE 9 — HUMAN FACTORS	LEVEL
	ALL
9.1 General The need to take human factors into account when performing maintenance; Incidents attributable to human factors/human error; Murphy's law.	2
9.2 Human performance and limitations Vision; Hearing; Information processing; Attention and perception; Memory; Claustrophobia and physical access.	2
9.3 Social psychology Accountability and responsibility: individual and group; Motivation and demotivation; Peer pressure; Cultural issues; Teamwork; Management, supervision, and leadership.	1
9.4 Factors that affect performance Fitness/health; Stress: domestic and work related; Time pressure and deadlines; Workload: overload, underload, and workload management; Sleep and fatigue, shift work; Alcohol, medication, drug abuse; Lack of manpower.	2
9.5 Physical environment Noise and fumes; Illumination; Climate and temperature; Motion and vibration; Working environment; Situational awareness.	1
9.6 Tasks Physical work; Repetitive tasks, complacency; Visual inspection; Complex systems; Critical maintenance tasks and error-capturing methods; Technical documentation: access, use, and quality.	1
9.7 Communication Within and between teams; Work logging and recording; Shift handover; Keeping up to date, currency; Dissemination of information.	2
9.8 Human error Error models and theories;	2

MODULE 9 — HUMAN FACTORS	LEVEL
	ALL
Types of error in maintenance tasks; Implications of errors (e.g. accidents); Organisational errors; Avoiding and managing errors.	
9.9 <i>Safety management</i> Risk management; Occurrence reporting; Safety culture Just culture; Identifying, avoiding, and reporting hazards; Organisational human-factors programme: professionalism and integrity, error-provoking behaviour, reporting errors, disciplinary policy, error investigation, action to address problems, feedback, assertiveness; Dealing with emergencies.	2
9.10 <i>The 'Dirty Dozen' and risk-mitigation</i> The 'Dirty Dozen': the twelve most common human-factors errors in maintenance: Lack of communication, Lack of teamwork, Lack of assertiveness, Complacency, Fatigue, Stress, Lack of knowledge, Lack of resources, Lack of awareness, Distraction, Pressure, Norms. Risk-mitigation methods.	2

MODULE 10 — AVIATION LEGISLATION

MODULE 10 — AVIATION LEGISLATION	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	B3
10.1 Regulatory framework Role of: — the International Civil Aviation Organization (ICAO); — the European Commission (EC); — the European Union Aviation Safety Agency (EASA); — the European Union Member States and national aviation authorities; — the bilateral agreements concluded by the European Commission; — Regulation (EU) 2018/1139 (the Basic Regulation) and its implementing acts: Regulations (EU) No 748/2012 (Initial Airworthiness) and (EU) No 1321/2014 (Continuing Airworthiness); — the relationship between regulations (hard law) and AMC, GM and CSs (soft law); — occurrence reporting according to Regulation (EU) No 376/2014; — the relationship between the various annexes (parts) relating to Initial and Continuing Airworthiness (such as Part 21, Part-M, Part-145, Part-66, Part-147, Part-T, Part-ML, Part-CAMO, and Part-CAO) and Regulations (EU) No 965/2012 (the Air Operations Regulation) and (EU) No 1178/2011 (the Air Crew Regulation).	1	1
10.2 Certifying staff — maintenance Deep understanding of Part-66 maintenance licences with the associated privileges and authorisations, and how to exercise them properly for the different aircraft categories.	2	2
10.3 Approved maintenance organisations General understanding of Part-145 and Part-CAO.	2	2
10.4 Independent certifying staff Privileges, responsibilities, record-keeping, limitations, and oversight according to Part-M, Part-66 and Part-ML.	—	3
10.5 Air operations General understanding of Regulation (EU) No 965/2012 (the Air Operations Regulation); Differences between commercial and non-commercial air operations, and their influence on aircraft maintenance; Air Operator Certificates (AOCs) and self-declaration authorisations; Air operator responsibilities, in particular regarding continuing airworthiness and maintenance; Specialised operations / specific approvals: ETOPS, CAT I/II/III, and BRNAV. Minimum Equipment List (MEL) and Configuration Deviation List (CDL); Aircraft placarding and markings; Documents to be carried on board: — Certificate of Airworthiness / Restricted Certificate of Airworthiness; — Airworthiness Review Certificate; — Permit to Fly; — Certificate of Registration; — Noise Certificate; — Weight and Balance report; — Radio Station Licence.	1	1

MODULE 10 — AVIATION LEGISLATION	LEVEL	
	A1	B1
	A2	B2
	A3	B2L
	A4	B3
10.6 <i>Certification of aircraft, parts, and appliances</i> Basic understanding of Part 21 and of the following EASA certification specifications: CS-22, CS-23, CS-25, CS-27, CS-29, and CS-STAN.	2	2
10.7 <i>Continuing airworthiness</i> General understanding of the Part 21 requirements on continuing airworthiness; General understanding of Part-M, Part-ML and Part-CAMO; Aircraft Maintenance Programme.	2	2
10.8 <i>Oversight principles in continuing airworthiness</i>	1	1
10.9 <i>Maintenance and certification beyond current EU regulations (if not superseded by EU requirements)</i> Maintenance of European Union aircraft that are not within the scope of Regulation (EU) 2018/1139 (Annex I aircraft); European military airworthiness requirement (EMAR) 66 licence; Applicable national and international requirements for component maintenance, welding, painting, NDT, etc. (if not superseded by EU requirements).	—	1
10.10 <i>Cybersecurity in aviation maintenance</i> Regulation on the introduction of organisation requirements for the management of information security risks related to aeronautical information systems used in civil aviation.	1	1

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.1 Theory of flight					
(a) Aeroplane aerodynamics and flight controls Operation and effect of: — roll control: ailerons and spoilers; — pitch control: elevators, stabilators, variable incidence stabilisers and canards; — yaw control, rudder limiters; — elevons, ruddervators; — high-lift devices, slots, slats, flaps, flaperons; — drag-inducing devices, spoilers, lift dumpers, speed brakes; — trim tabs, servo tabs, control surface bias.	1	1	2	2	1
(b) Aeroplane: other aerodynamic devices Operation and effect of: — balance and antibalance (leading) tabs; — spring tabs, mass balance, aerodynamic balance panels; — mass balance, aerodynamic balance panels; — effects of wing fences, saw tooth leading edges; — boundary layer control using vortex generators, stall wedges or leading-edge devices.	1	1	2	2	1
11.2 Airframe structures (ATA 51)					
(a) General concepts: — Zonal and station identification systems; — Electrical bonding; — Lightning strike protection provisions.	2	2	2	2	2
(b) Airworthiness requirements for structural strength: — Structural classification: primary, secondary, and tertiary; — Fail-safe, safe-life, damage-tolerance concepts; — Stress, strain, bending, compression, shear, torsion, tension, hoop stress, fatigue; — Drains and ventilation provisions; — System installation provisions.	2	2	2	2	2
(c) Construction methods — Stressed skin fuselage, formers, stringers, longerons, bulkheads, frames, doublers, struts, ties, beams, floor structures, reinforcement, skinning, anticorrosive protection, wing, empennage and engine attachments; — Structure assembly techniques: riveting, bolting, bonding; — Methods of surface protection, such as chromating, anodising, painting; — Surface cleaning; — Airframe symmetry: methods of alignment and symmetry checks.	1	1	2	2	2

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.3 <i>Airframe structures — aeroplanes</i>					
11.3.1 <i>Fuselage, doors, windows (ATA 52/53/56)</i>	1	1	2	2	1
(a) Construction principles					
— Construction and pressurisation sealing;					
— Wing, stabiliser, pylon, and undercarriage attachments;					
— Seat installation and cargo loading system;					
— Doors and emergency exits: construction, mechanisms, operation and safety devices;					
— Windows and windscreen construction and mechanisms.					
(b) Airborne towing devices (glider, banner, target).	1	1	1	1	1
(c) Doors	1	1	2	1	—
— Doors and emergency exits: safety devices;					
— Cargo loading system.					
11.3.2 <i>Wings (ATA 57)</i>	1	1	2	2	1
Construction;					
Fuel storage;					
Landing gear, pylon, control surface and high lift/drag attachments.					
11.3.3 <i>Stabilisers (ATA 55)</i>	1	1	2	2	1
Construction;					
Control surface attachment.					
11.3.4 <i>Flight control surfaces (ATA 55/57)</i>	1	1	2	2	1
Construction and attachments;					
Balancing — mass and aerodynamics.					
11.3.5 <i>Nacelles/pylons (ATA 54)</i>	1	1	2	2	1
Nacelles/Pylons:					
— Construction,					
— Firewalls,					
— Engine mounts.					
11.4 <i>Air conditioning and cabin pressurisation (ATA 21)</i>					
(a) Pressurisation	1	1	3	3	—
Pressurisation systems;					
Cabin pressure controllers, control, and safety valves;					
Control and indication.					
(b) Air supply	1	-	3	-	—
Sources of air supply including engine bleed, APU and ground cart;					
Distribution systems.					
(c) Air conditioning	1	—	3	-	—
Air-conditioning systems;					
Air cycle and vapour cycle machines;					
Flow, temperature and humidity control system;					
Control and indication control valves.					
(d) Safety and warning devices	1	1	3	3	—
Protection and warning devices.					
(e) Heating and ventilation systems.	—	1	-	3	1

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.5 Instruments / avionics systems					
11.5.1 Instrument systems (ATA 31)	1	1	2	2	2
Pitot-static: Airspeed indicators, Vertical speed indicators, Altimeters; Gyroscopic: Gyroscopic principles, Artificial horizons, Attitude directors, Direction indicators, Horizontal situation indicators (HSI), Slip indicators, Turn indicators, Turn coordinators; Compass systems: systems, direct reading, remote reading, Stall-warning systems and angle-of-attack indicating systems, Glass cockpit, Indications of other aircraft systems.					
11.5.2 Avionics systems	1	1	1	1	1
Fundamentals of system layouts and operation of: Autoflight (ATA 22) ; Communication systems (ATA 23) : — Very High Frequency (VHF) communications, — High Frequency (HF) communications, — Satellite Communications (SATCOM), — Controller–pilot data link communications (CPDLC), — Audio systems, — Emergency Locator Transmitters (ELTs), — Cockpit Voice Recorder (CVR); Navigation systems (ATA 34) : — Very high frequency omnidirectional range (VOR), — Automatic direction finder (ADF), — Instrument landing system (ILS), — Microwave landing system (MLS), — Flight director systems (FDSs), distance-measuring equipment (DME), — Area navigation (RNAV) systems, — Flight management systems (FMSs), — Satellite navigation systems, — Air traffic control transponder, secondary surveillance radar, — Traffic alert and collision avoidance system (TCAS), — Weather avoidance radar, — Radio altimeter, — Inertial navigation system (INS), — ARINC (Aeronautical Radio Incorporated) communication and reporting. Types and uses of avionics general test equipment.					

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
11.6 Electrical power (ATA 24) — Installation and operation of batteries; — DC power generation; — AC power generation; — Emergency power generation; — Voltage regulation; — Power distribution; — Inverters, transformers, rectifiers; — Circuit protection; — External/ground power.	1	1	3	3	3
11.7 Equipment and furnishings (ATA 25) (a) Emergency equipment: Emergency equipment requirements. (b) Cabin and cargo layout: — Seats, harnesses, and belts; — Cabin layout; — Equipment layout; — Cabin furnishing installation; — Galley installation; — Cargo handling and retention equipment; — Airstairs.	2	2	2	2	2
11.8 Fire protection (ATA 26) (a) Fire and smoke detection system, and fire-extinguishing systems: — Fire and smoke detection and warning systems; — Fire-extinguishing systems; — System tests. (b) Portable fire extinguisher.	1	1	1	1	—
	1	1	1	1	1
11.9 Flight controls (ATA 27) (a) Primary and secondary flight controls: — Primary controls: aileron, elevator, rudder, spoiler; — Trim control, trim tabs; — High-lift devices; — System operation: manual; — Gust locks and gust lock systems; — Artificial feel, yaw damper, Mach trim, rudder limiter; — Stall-warning systems. (b) Actuation and protection: — Active load control; — Lift dump, speed brakes; — Hydraulic, pneumatic systems; — Stall-protection systems. (c) System operation: Electrical systems, fly-by-wire systems. (d) Balancing and rigging.	1	1	3	3	2
	1	—	3	—	—
	1	—	3	—	—
	1	1	3	3	2
11.10 Fuel systems (ATA 28, ATA 47)					

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
(a) Systems: — System layout; — Fuel tanks; — Supply systems.	1	1	3	3	1
(b) Fuel handling: — Cross-feed and transfer; — Refuelling and defuelling.	1	1	3	3	1
(c) Indication and warnings.	1	1	3	3	1
(d) Special systems: — Dumping, venting, and draining; — Inert gas systems.	1	—	3	—	—
(e) Balancing: Longitudinal balance fuel systems.	1	—	3	—	—
11.11 Hydraulic power (ATA 29)					
(a) System description: System layout; Hydraulic fluids; Hydraulic reservoirs and accumulators; Filters; Power distribution.	1	1	3	3	2
(b) System operation (1): Pressure generation: electric and mechanical; Pressure control; Indication and warning systems; Servicing.	1	1	3	3	2
(c) System operation (2): Pressure generation: pneumatic; Emergency pressure generation; Interface with other systems.	1	—	3	—	—
11.12 Ice and rain protection (ATA 30)					
(a) Principles: Ice formation, classification, and detection.	1	1	3	3	1
(b) De-icing: De-icing systems: electrical, hot-air, pneumatic, chemical; Probe and drain heating.	1	1	3	3	1
(c) Anti-icing: Anti-icing systems: electrical, hot-air, chemical.	1	—	3	—	—
(d) Wipers: Wiper systems.	1	1	3	3	1
(e) Rain-repellent systems.	1	—	3	—	—
11.13 Landing gear (ATA 32)					
(a) Description: Construction, shock absorbing; Tyres.	2	2	3	3	2

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
(b) Systems: Extension and retraction systems: normal and emergency; Indications and warnings; Wheels, brakes, antiskid, and autobraking; Steering.	2	2	3	3	2
(c) Air–ground sensing.	2	—	3	—	—
(d) Tail protection: Skids.	2	2	3	3	2
11.14 Lights (ATA 33) External: navigation, anticollision, landing, taxiing, ice; Internal: cabin, cockpit, cargo; Emergency.	2	2	3	3	2
11.15 Oxygen (ATA 35) System layout: cockpit, cabin; Sources, storage, charging and distribution; Supply regulation; Indications and warnings.	1	1	3	3	2
11.16 Pneumatic / vacuum (ATA 36) (a) Systems: System layout; Sources: engine / APU (Auxiliary Power Unit), compressors, reservoirs, ground supply; Pressure control; Distribution; Indications and warnings; Interface with other systems. (b) Pumps: Pressure and vacuum pumps.	1	1	3	3	2
	1	1	3	3	2
11.17 Water / waste (ATA 38) (a) Systems: Water system layout, supply, distribution, servicing and draining; Toilet system layout, flushing and servicing. (b) Corrosion: Corrosion aspects.	2	2	3	3	2
	2	2	3	3	2
11.18 Onboard maintenance systems (ATA 45) Central maintenance computers; Data-loading system; Electronic library system; Printing systems; Structure monitoring (damage-tolerance monitoring).	1	—	2	—	—
11.19 Integrated modular avionics (IMA) (ATA 42) (a) Overall system description and theory: Core system; network components; Functions that may be typically integrated in the integrated modular avionics (IMA) modules are, among others:	1	—	2	—	—

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL				
	A1	A2	B1.1	B1.2	B3
Bleed management, air pressure control, air ventilation and control, avionics and cockpit ventilation control, temperature control, air traffic communication, avionics communication router, electrical load management, circuit breaker monitoring, electrical system BITE, fuel management, braking control, steering control, landing gear extension and retraction, tyre pressure indication, oleo pressure indication, brake temperature monitoring, etc.					
(b) Typical system layout.	1	—	2	—	—
11.20 Cabin systems (ATA 44)	1	—	2	—	—
System architecture, operation, and control of systems for: — passenger in-flight entertainment; — communication within the aircraft (Cabin intercommunication data system (CIDS); — communication between the aircraft cabin and ground stations; — including voice, data, music, and video transmission. CIDS interface between cockpit/cabin crew and cabin systems. Data exchange between the different related line replaceable units (LRUs). Flight attendant panels (FAPs). Cabin network server (CNS) and interfaces with the following systems: — Data/radio communication; — Cabin core system (CCS); — In-flight entertainment system (IFES); — External communication system (ECS); — Cabin mass memory system (CMMS); — Cabin monitoring system (CMS); — Miscellaneous cabin systems (MCSs); and — Other systems. Cabin network server (CNS) hosting functions: — Access to predeparture/departure reports; — Email/intranet/internet access; passenger database; — In-flight entertainment system; — External communication system; — Cabin mass memory system; — Cabin monitoring system; — Miscellaneous cabin system.					
11.21 Information systems (ATA 46)	1	—	2	—	—
System architecture, operation, and control of: — Storage and electronic library; — Updating; — Retrieving of digital information; — Air traffic and information management systems (ATIMS) and network server systems; — Aircraft general information system; — Flight deck information system; — Maintenance information system; — Passenger cabin information system; — Miscellaneous information systems; — Other linked systems.					

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
12.1 Theory of flight — rotary wing aerodynamics Terminology; Effects of gyroscopic precession; Torque reaction and directional control; Dissymmetry of lift, blade tip stall; Translating tendency and its correction; Coriolis effect and compensation; Vortex ring state, power setting, overpitching; Auto-rotation; Ground effect.	1	2
12.2 Flight control systems (ATA 67) Cyclic control; Collective control; Swashplate; Yaw control: antitorque control, tail rotor, bleed air; Main-rotor head: design and operation features; Blade dampers: function and construction; Rotor blades: main- and tail-rotor blade construction and attachment; Trim control, fixed and adjustable stabilisers; System operation: manual, hydraulic, electrical, fly-by-wire; Artificial feel; Balancing and rigging.	2	3
12.3 Blade tracking and vibration analysis (ATA 18) Rotor alignment; Main-rotor and tail-rotor tracking; Static and dynamic balancing; Vibration types, and vibration reduction methods; Ground resonance.	1	3
12.4 Transmission Gear boxes, main and tail rotors; Clutches, free wheel units and rotor brake; Tail-rotor drive shafts, flexible couplings, bearings, vibration dampers and bearing hangers.	1	3
12.5 Airframe structures (a) General concept Airworthiness requirements for structural strength; Structural classification: primary, secondary, tertiary; Fail-safe, safe-life, damage-tolerance concepts; Zonal and station identification systems; Stress, strain, bending, compression, shear, torsion, tension, hoop stress, fatigue; Drains and ventilation provisions; System installation provisions; Lightning strike protection provisions.	2	2

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
(b) Construction methods for the principal elements Stressed skin fuselage, formers, stringers, longerons, bulkheads, frames, doublers, struts, ties, beams, floor structures, reinforcement, skinning and anticorrosive protection; Pylon, stabiliser and undercarriage attachments; Seat installation; Doors: construction, mechanisms, operation, and safety devices; Windows and windscreen construction; Fuel storage; Firewalls; Engine mounts; Structure assembly techniques: riveting, bolting, bonding; Methods for surface protection, such as chromating, anodising, painting; Surface cleaning; Airframe symmetry: methods for alignment and symmetry checks.	1	2
12.6 Air conditioning (ATA 21) 12.6.1 Air supply Sources of air supply, including engine bleed and ground cart.	1	2
12.6.2 Air conditioning Air-conditioning systems; Distribution systems; Flow and temperature control systems; Protection and warning devices.	1	3
12.7 Instruments / avionics systems 12.7.1 Instrument systems (ATA 31) Pitot-static: altimeter, airspeed indicator, vertical speed indicator; Gyroscopic: artificial horizon, attitude director, direction indicator, horizontal situation indicator, turn and slip indicator, turn coordinator; Compasses: direct reading, remote reading; Vibration indicating systems / health and usage monitoring systems (HUMSs); Glass cockpit; Indications of other aircraft systems.	1	2
12.7.2 Avionics systems Fundamentals of system layouts and operation of: Autoflight (ATA 22); Communications (ATA 23): — Very high frequency (VHF) communications, — High-frequency (HF) communications, — Satellite communications (SATCOM), — Controller–pilot data link communications (CPDLC), — Audio systems, — Emergency locator transmitters (ELTs), — Cockpit voice recorder (CVR); Navigation systems (ATA 34):	1	1

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
— Very high frequency omnidirectional range (VOR), — Automatic direction finding (ADF), — Instrument landing system (ILS), — Microwave landing system (MLS), — Flight director systems (FDSs), distance-measuring equipment (DME), — Area navigation (RNAV) systems, — Flight management systems (FMSs), — Satellite navigation systems, — Inertial navigation system (INS), — Air traffic control transponder, secondary surveillance radar, — Traffic alert and collision avoidance system (TCAS), — Weather avoidance radar, — Radio altimeter, — ARINC communication and reporting. Types and uses of general test equipment for avionics.		
12.8 Electric power (ATA 24) Installation and operation of batteries; DC power generation, AC power generation; Emergency power generation; Voltage regulation, circuit protection; Power distribution; Inverters, transformers, rectifiers; External/Ground power.	1	3
12.9 Equipment and furnishings (ATA 25) (a) Emergency equipment requirements; Seats, harnesses, and belts; Lifting systems. (b) Emergency flotation systems; Cabin layout, cargo retention; Equipment layout; Cabin furnishing installation.	2	2
	1	1
12.10 Fire protection (ATA 26) (a) Fire and smoke detection and warning systems; Fire-extinguishing systems; System tests. (b) Portable fire extinguishers.	1	3
	1	1
12.11 Fuel systems (ATA 28) System layout; Fuel tanks; Supply systems; Dumping, venting, and draining; Cross-feed and transfer; Indications and warnings; Refuelling and defuelling.	1	3

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
12.12 Hydraulic power (ATA 29) System layout; Hydraulic fluids; Hydraulic reservoirs and accumulators; Pressure generation: electric, mechanical, pneumatic; Emergency pressure generation; Filters; Pressure control; Power distribution; Indication and warning systems; Interface with other systems; Servicing.	1	3
12.13 Ice and rain protection (ATA 30) Ice formation, classification, and detection; Anti-icing and de-icing systems: electrical, hot-air, and chemical; Rain repellent and removal; Probe and drain heating; Wiper system.	1	3
12.14 Landing gear (ATA 32) (a) System description and operation: Construction, shock absorbing; Extension and retraction systems: normal and emergency; Wheels, tyres, brakes; Steering; Skids, floats. (b) Sensors: Indications and warning; Air-ground sensing.	2	3
	2	3
12.15 Lights (ATA 33) External: navigation, landing, taxiing; Internal: cabin, cockpit, cargo; Emergency.	2	3
12.16 (Reserved)		
12.17 Integrated modular avionics (IMA) (ATA 42)	1	2

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL	
	A3 A4	B1.3 B1.4
(a) Overall system description and theory Functions that may be typically integrated in the integrated modular avionics (IMA) modules: Bleed management, air pressure control, air ventilation and control, avionics and cockpit ventilation control, temperature control, air traffic communication, avionics communication router, electrical load management, circuit breaker monitoring, electrical system BITE, fuel management, steering control, landing gear extension and retraction, tyre pressure indication, oleo pressure indication, brake temperature monitoring, etc.; Core System; Network Components.		
(b) Typical system layouts.	1	2
12.18 Onboard maintenance systems (ATA 45) Central maintenance computers; Data-loading system; Electronic library system.	1	2
12.19 Information systems (ATA 46) The units and components which furnish a means of storing, updating, and retrieving digital information traditionally provided on paper, microfilm or microfiche. They include units that are dedicated to the information storage and retrieval function, such as the electronic library mass storage and controller. They do not include units or components installed for other uses and shared with other systems, such as flight deck printer or general-use display. Typical examples include air traffic and information management systems and network server systems. Aircraft general information system. Flight deck information system. Maintenance information system. Miscellaneous information system.	1	2

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS

C/N: Communication and Navigation; **Ins.:** Instruments; **A/F:** Autoflight; **Sur.:** Surveillance; **A/S:** Airframe and Systems

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
13.1 Theory of flight							
(a) Aeroplane aerodynamics and flight controls Operation and effect of: — roll control: ailerons and spoilers; — pitch control: elevators, stabilators, variable incidence stabilisers and canards; and — yaw control: rudder limiters; — control using elevons, ruddervators; — high-lift devices: slots, slats, flaps; — drag-inducing devices: spoilers, lift dumpers, speed brakes; — trim tabs, servo tabs, and control surface bias.	1	1	—	—	—	—	—
(b) Rotary wing aerodynamics Terminology; Operation and effect of cyclic, collective, and antitorque controls.	1	1	—	—	—	—	—
13.2 Structures — general concepts							
(a) General concepts; Zonal and station identification systems; Electrical bonding; Lightning strike protection provisions.	2	2	—	—	—	—	—
(b) Fundamentals of structural systems.	1	1	—	—	—	—	—
13.3 Autoflight (ATA 22)							
(a) Fundamentals of automatic flight control: — Working principles and current terminology; — Command signal processing; — Modes of operation: roll, pitch, and yaw channels; — Yaw dampers; — Stability augmentation system in helicopters; — Automatic trim control; — Autopilot navigation aids interface.	3	—	—	—	3	—	—
(b) Autothrottle systems and automatic landing systems: — Principles and categories; — Modes of operation; — Approach; — Glideslope; — Land, go-around; — System monitors and failure conditions.	3	—	—	—	3	—	—

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
13.4 Communication/navigation (ATA 23/34)							
(a) Fundamentals of communication and navigation systems:	3	—	3	—	—	—	—
— Radio wave propagation, antennas, transmission lines, communication, receiver, and transmitter;							
Working principles of the following systems:							
— Very high frequency (VHF) communications;							
— High-frequency (HF) communications;							
— Satellite communications (SATCOM);							
— Controller–pilot data link communications (CPDLC);							
— Audio systems;							
— Emergency locator transmitters (ELTs);							
— Cockpit voice recorder (CVR);							
— Very high frequency omnidirectional range (VOR);							
— Automatic direction finding (ADF);							
— Instrument landing system (ILS);							
— Flight director systems (FDSs), distance-measuring equipment (DME);							
— Area navigation (RNAV) systems;							
— Flight management systems (FMSs);							
— Global navigation satellite systems (GNSSs), Global Positioning System (GPS), ground-based augmentation system (GBAS), satellite-based augmentation system (SBAS) such as the European geostationary navigation overlay service (EGNOS) and the wide area augmentation system (WAAS);							
— Data link and two-way data link.							
(b) Fundamentals of aircraft surveillance systems:	3	—	—	—	—	3	—
— Air traffic control transponder, secondary surveillance radar;							
— Traffic alert and collision avoidance system (TCAS);							
— Weather avoidance radar;							
— Radio altimeter;							
— Automatic dependent surveillance — broadcast (ADS-B) and its other associated services such as FIS-B, TIS-B and multilink;							
— Inertial navigation system (INS);							
— ARINC (Aeronautical Radio Incorporated) communication and reporting.							
13.5 Electric power (ATA 24)							

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
— Installation and operation of batteries; — DC power generation; — AC power generation; — Emergency power generation; — Voltage regulation; — Power distribution; — Inverters, transformers, rectifiers; — Circuit protection; — External/ground power.	3	3	—	—	—	—	—
13.6 Equipment and furnishings (ATA 25) Electronic emergency equipment requirements.	3	—	—	—	—	—	—
13.7 Flight controls (a) Primary and secondary flight controls (ATA 27): — Primary controls: aileron, elevator, rudder, spoiler; — Trim control: trim tabs; — High-lift devices; — System operation: manual; — Gust locks and gust lock systems; — Artificial feel, yaw damper, Mach trim, rudder limiter; — Stall-warning systems.	2	—	—	—	2	—	—
(b) Actuation and protection: — Active load control; — Lift dump, speed brakes; — Hydraulic, pneumatic systems; — Stall-protection systems.	2	—	—	—	2	—	—
(c) System operation: — System operation: electrical, fly-by-wire.	3	—	—	—	3	—	—
(d) Rotorcraft flight controls (ATA 67): Rotorcraft controls: cyclic control, collective control, swashplate, yaw control.	2	—	—	—	2	—	—
13.8 Instruments (ATA 31) — Classification; — Atmosphere; — Terminology; — Pressure-measuring devices and systems; — Pitot-static systems; — Altimeters; — Vertical speed indicators; — Airspeed indicators; — Machmeters; — Altitude-reporting/-alerting systems; — Air-data computers; — Instrument pneumatic systems; — Direct-reading pressure and temperature gauges; — Temperature-indicating systems;	3	—	—	3	—	—	—

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
— Gyroscopic principles; — Artificial horizons; — Slip indicators; — Directional gyros; — Ground proximity warning systems (GPWSs); — Compass systems; — Flight data recording systems (FDRs); — Electronic flight instrument systems (EFISs) — typical system arrangements and cockpit layout; — Instrument warning systems, including master warning systems and centralised warning panels; — Stall-warning systems and angle-of-attack indicating systems; — Vibration measurement and indication; — Glass cockpit; — Types and uses of general test equipment for avionics.							
13.9 Lights (ATA 33) External: navigation, anticollision, landing, taxiing, ice; Internal: cabin, cockpit, cargo; Emergency.	3	3	—	—	—	—	—
13.10 Onboard maintenance systems (ATA 45) Central maintenance computers; Data-loading system; Electronic library system; Printing system; Structure monitoring system (damage-tolerance monitoring).	3	—	—	—	—	—	—
13.11 Air conditioning and cabin pressurisation (ATA 21) (a) Pressurisation: — Pressurisation systems; — Cabin pressure controllers, control and safety valves; — Control and indication. (b) Air supply: Sources of air supply including engine bleed, APU and ground cart; Distribution systems.	3	—	—	—	—	—	3
(c) Air conditioning.	1	—	—	—	—	—	1
(d) Safety and warning devices.	3	—	—	—	—	—	3
13.12 Fire protection (ATA 26) (a) Fire and smoke detection system and fire-extinguishing systems; — Fire and smoke detection and warning systems; — Fire-extinguishing systems; — System tests. (b) Portable fire extinguisher.	3	—	—	—	—	—	3
	1	—	—	—	—	—	1

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
13.13 Fuel systems (ATA 28, ATA 47)							
(a) System layout: System layout; Fuel tanks; Supply systems.	1	—	—	—	—	—	1
(b) Fuel handling; Cross-feed and transfer; Refuelling and defuelling.	2	—	—	—	—	—	2
(c) Indications and warnings.	3	—	—	—	—	—	3
(d) Special systems: Dumping, venting and draining; Inert gas systems.	1	—	—	—	—	—	1
(e) Balancing: Longitudinal balance fuel systems.	3	—	—	—	—	—	3
13.14 Hydraulic power (ATA 29)							
(a) System layout: System layout; Hydraulic fluids; Hydraulic reservoirs and accumulators; Filters; Power distribution.	1	—	—	—	—	—	1
(b) System operation (1): Pressure generation: electric and mechanical; Pressure control; Indication and warning systems; Servicing.	3	—	—	—	—	—	3
(c) System operation (2): Pressure generation: pneumatic; Emergency pressure generation; Interface with other systems.	3	—	—	—	—	—	3
13.15 Ice and rain protection (ATA 30)							
(a) Principles: Ice formation, classification, and detection.	2	—	—	—	—	—	2
(b) De-icing: De-icing systems: electrical, hot-air, pneumatic, and chemical; Probe and drain heating.	3	—	—	—	—	—	3
(c) Anti-icing: Anti-icing systems: electrical, hot-air, and chemical.	2	—	—	—	—	—	2
(d) Wiper systems.	1	—	—	—	—	—	1
(e) Rain repellent.	1	—	—	—	—	—	1
13.16 Landing gear (ATA 32)							
(a) Description: Construction, shock absorbing; Tyres.	1	—	—	—	—	—	1
(b) Systems: Extension and retraction systems: normal and emergency; Indications and warnings;	3	—	—	—	—	—	3

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MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	LEVEL						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
System architecture, operation and control of systems for: — passenger in-flight entertainment; — communication within the aircraft (cabin intercommunication data system (CIDS); — communication between the aircraft cabin and ground stations, including voice, data, music and video transmission. CIDS interface between cockpit/cabin crew and cabin systems; - Data exchange between the different related line replaceable units (LRUs); - Flight attendant panels (FAPs). CNS server and interfaces with the following systems: — Data/radio communication system; — Cabin core system (CCS); — In-flight entertainment system (IFES); — External communication system (ECS); — Cabin mass memory system (CMMS); — Cabin monitoring system (CMS); — Miscellaneous cabin systems (MCSS). The CNS may host functions such as: — access to predeparture/departure reports; — email/intranet/internet access; — passenger database.	3	—	—	—	—	—	—
13.22 Information systems (ATA 46) The units and components which furnish a means of storing, updating, and retrieving digital information traditionally provided on paper, microfilm or microfiche. They include units that are dedicated to the information storage and retrieval function, such as the electronic library mass storage and controller, but they do not include units or components installed for other uses and shared with other systems, such as flight deck printer or general-use display. Typical examples include: — air traffic and information management systems and network server systems; — aircraft general information system; — flight deck information system; — maintenance information system; — passenger cabin information system; — miscellaneous information systems; — other linked systems.	3	—	—	—	—	—	—

MODULE 14 — PROPULSION

MODULE 14 — PROPULSION	LEVEL
	B2 B2L Instruments B2L Airframe & Systems
14.1 Engines	
(a) Constructional arrangement and operation of turbojet, turbofan, turboshaft, and turboprop engines.	1
(b) Constructional arrangement and operation of auxiliary power units (APUs).	1
(c) Constructional arrangement and operation of piston engines.	1
(d) Constructional arrangement and operation of electric and hybrid engines, their electric energy storage and control systems.	2
(e) Electronic engine control and fuel-metering systems (full authority digital engine control (FADEC)).	2
14.2 Electric/electronic engine indication systems	2
— Exhaust gas temperature / interstage turbine temperature systems; — Cylinder head temperature, engine coolant temperature, engine speed; — Engine thrust indication: engine pressure ratio, engine turbine discharge pressure or jet pipe pressure systems; — Vibration measurement systems; — Oil pressure and temperature; — Fuel pressure, temperature, and flow; — Manifold pressure; — Engine torque.	
14.3 Propeller systems	2
— Propeller speed indication; — Speed control and pitch change methods — electrical/electronic; — Synchronising and synchrophasing equipment; — Electrical anti-icing/de-icing equipment.	
14.4 Starting and ignition systems	2
— Operation of engine start systems and components; — Ignition systems and components; — Maintenance safety requirements.	

MODULE 15 — GAS-TURBINE ENGINE

MODULE 15 — GAS-TURBINE ENGINE	LEVEL	
	A1 A3	B1.1 B1.3
15.1 Fundamentals — Potential energy, kinetic energy, Newton's laws of motion, Brayton cycle; — The relationship between force, work, power, energy, velocity, and acceleration; — Constructional arrangement and operation of turbojet, turbofan, turboshaft, turboprop, and geared turbofan engines.	1	2
15.2 Engine performance — Gross thrust, net thrust, choked nozzle thrust, thrust distribution, resultant thrust, thrust horsepower, equivalent shaft horsepower, specific fuel consumption; — Engine efficiencies; — By-pass ratio and engine pressure ratio; — Pressure, temperature, and velocity of the gas flow; — Engine ratings, static thrust, influence of speed, altitude and hot climate, flat rating, limitations.	—	2
15.3 Inlet — Compressor inlet ducts; — Effects of various inlet configurations; — Ice protection.	2	2
15.4 Compressors — Axial and centrifugal types; — Constructional features, operating principles, and applications; — Fan balancing; — Operation: - o Causes and effects of compressor stall and surge; - o Methods of air-flow control: bleed valves, variable inlet guide vanes, variable stator vanes, rotating stator blades; - o Compressor ratio.	1	2
15.5 Combustion section Constructional features and principles of operation.	1	2
15.6 Turbine section — Operation and characteristics of different turbine blade types; — Blade-to-disk attachment; — Nozzle guide vanes; — Causes and effects of turbine blade stress and creep.	2	2
15.7 Exhaust — Constructional features and principles of operation; — Convergent, divergent, and variable area nozzles; — Engine noise reduction; — Thrust reversers.	1	2
15.8 Bearings and seals Constructional features and principles of operation.	—	2
15.9 Lubricants and fuels — Properties and specifications of standard, alternate, and drop-in fuel; — Properties and specifications of lubricants; — Fuel additives; — Safety precautions.	1	2

MODULE 15 — GAS-TURBINE ENGINE	LEVEL	
	A1 A3	B1.1 B1.3
15.10 <i>Lubrication systems</i> System operation/layout and components.	1	2
15.11 <i>Fuel systems</i> — Operation of engine control and fuel-metering systems, including electronic engine control (full authority digital engine control (FADEC)) and electronic power augmentation; — System layout and components.	1	2
15.12 <i>Air systems</i> Operation of engine air distribution and anti-icing control systems, including internal cooling and sealing, and external air services.	1	2
15.13 <i>Starting and ignition systems</i> — Operation of engine start systems and components; — Ignition systems and components; — Maintenance safety requirements.	1	2
15.14 <i>Engine indication systems</i> — Exhaust gas temperature / interstage turbine temperature; — Engine thrust indication: engine pressure ratio, engine turbine discharge pressure or jet pipe pressure systems; — Oil pressure and temperature; — Fuel pressure and flow; — Engine speed; — Vibration measurement and indication; — Torque; — Power.	1	2
15.15 <i>Alternate turbine constructions</i> — Geared turbofan (GTF); — Variable fan blades; — Open rotor/propfan; — Hybrid turbine-electric concepts and electric power augmentation; — Future trends and developments.	—	1
15.16 <i>Turboprop engines</i> — Gas-coupled/free-turbine and gear-coupled turbines; — Reduction gears; — Integrated engine and propeller controls; — Overspeed safety devices.	1	2
15.17 <i>Turboshaft engines</i> Arrangements, drive systems, reduction gearing, couplings, control systems.	1	2
15.18 <i>Auxiliary power units (APUs)</i> Purpose, operation, protective systems.	1	2
15.19 <i>Power plant installation</i> Configuration of firewalls, cowlings, acoustic panels, engine mounts, antivibration mounts, hoses, pipes, feeders, connectors, wiring looms, control cables and rods, lifting points and drains.	1	2
15.20 <i>Fire protection systems</i> Operation of fire-detection and fire-extinguishing systems.	1	2

MODULE 15 — GAS-TURBINE ENGINE	LEVEL	
	A1 A3	B1.1 B1.3
15.21 Engine monitoring and ground operation — Procedures for starting and ground run-up; — Interpretation of engine power output and parameters; — Trend (including oil analysis, vibration and borescope) monitoring; — Inspection of engine and components to criteria, tolerances and data specified by the engine manufacturer; — Compressor washing/cleaning; — Foreign object damage (FOD).	1	3
15.22 Engine storage and preservation Preservation and depreservation for the engine and its accessories/systems.	—	2

MODULE 16 — PISTON ENGINE

MODULE 16 — PISTON ENGINE	LEVEL	
	A2 A4	B1.2 B1.4 B3
16.1 Fundamentals — Mechanical, thermal, and volumetric efficiencies; — Operating principles: 2-stroke, 4-stroke, Otto, diesel, and rotary (Wankel); — Piston displacement and compression ratio; — Engine configuration and firing order.	1	2
16.2 Engine performance — Power calculation and measurement; — Factors that affect engine power; — Mixtures/leaning, pre-ignition.	1	2
16.3 Engine construction — Crank case, crank shaft, cam shafts, sumps; — Accessory gearbox; — Cylinder and piston assemblies; — Connecting rods, inlet and exhaust manifolds; — Valve mechanisms; — Propeller reduction gearboxes.	1	2
16.4 Engine fuel systems		
16.4.1 Carburettors — Types, construction, and principles of operation; — Icing and heating.	1	2
16.4.2 Fuel injection systems Types, construction, and principles of operation.	1	2
16.4.3 Electronic engine control — Operation of engine control and fuel-metering systems including electronic engine control (full authority digital engine control (FADEC)); — System layout and components.	1	2
16.5 Starting and ignition systems — Starting systems, preheat systems; — Magneto types, construction, and principles of operation; — Ignition harnesses, spark plugs; — Low- and high-tension systems.	1	2
16.6 Induction, exhaust, and cooling systems — Construction and operation of induction systems, including alternate air systems; — Exhaust systems, engine cooling systems — air and liquid.	1	2
16.7 Supercharging/turbocharging — Principles and purpose of supercharging and its effects on engine parameters; — Construction and operation of supercharging/turbocharging systems; — System terminology; — Control systems; — System protection.	1	2
16.8 Lubricants and fuels — Properties and specifications of standard, alternate, and drop-in fuel; — Properties and specifications of lubricants; — Fuel additives; — Safety precautions.	1	2

MODULE 16 — PISTON ENGINE	LEVEL	
	A2 A4	B1.2 B1.4 B3
16.9 <i>Lubrication systems</i> System operation/layout and components.	1	2
16.10 <i>Engine indication systems</i> — Engine speed; — Cylinder head temperature; — Coolant temperature; — Oil pressure and temperature; — Exhaust gas temperature; — Fuel pressure and flow; — Manifold pressure.	1	2
16.11 <i>Power plant installation</i> Configuration of firewalls, cowlings, acoustic panels, engine mounts, antivibration mounts, hoses, pipes, feeders, connectors, wiring looms, control cables and rods, lifting points and drains.	1	2
16.12 <i>Engine monitoring and ground operation</i> — Procedures for starting and ground run-up; — Interpretation of engine power output and parameters; — Inspection of engine and components: criteria, tolerances, and data specified by the engine manufacturer.	1	3
16.13 <i>Engine storage and preservation</i> Preservation and depreservation for the engine and its accessories/systems.	—	2
16.14 <i>Alternate piston-engine constructions</i> Hybrid piston–electric concepts and electric power augmentation.	1	1

MODULE 17 — PROPELLER

MODULE 17 — PROPELLER	LEVEL	
	A1 A2	B1.1 B1.2 B3
17.1 Fundamentals — Blade element theory; — High/low blade angle, reverse angle, angle of attack, rotational speed; — Propeller slip; — Aerodynamic, centrifugal, and thrust forces; — Torque; — Relative airflow on blade angle of attack; — Vibration and resonance.	1	2
17.2 Propeller construction — Construction methods and materials used in wooden, composite and metal propellers; — Blade station, blade face, blade shank, blade back / thrust face and hub assembly; — Fixed pitch, controllable pitch, constant speeding propeller; — Propeller/spinner installation.	1	2
17.3 Propeller pitch control — Speed control and pitch change methods — mechanical and electrical/electronic; — Feathering and reverse pitch; — Overspeed protection.	1	2
17.4 Propeller synchronising Synchronising and synchrophasing equipment.	—	2
17.5 Propeller ice protection Fluid and electrical de-icing equipment.	1	2
17.6 Propeller maintenance — Static and dynamic balancing; — Blade tracking; — Assessment of blade damage, erosion, corrosion, impact damage, delamination; — Propeller treatment/repair schemes; — Propeller engine running.	1	3
17.7 Propeller storage and preservation Propeller preservation and depreservation.	1	2

AMC1 Appendix I — Basic Knowledge Requirements (except for category L licence) Section 3

ED Decision 2023/019/R

Basic training methods

Training methods are categorised as follows: ‘instructor-centred training’, ‘student-centred training’, and ‘blended training’.

The actual training method and training tools should be adapted to suit the training subject and be chosen considering their intrinsic characteristics, such as but not limited to their efficiency and the pedagogical benefits of the training method/tool.

Basic training modules 7, 9, 11, 12, 13, 14, 15, 16 and 17 should not normally be taught solely through a student-centred method unless provisions are in place to verify the actual and progressive acquisition of knowledge, skills and attitude by the student.

Appendix II — Basic examination standard (except for category L licence)

1. General

Regulation (EU) 2023/989

- 1.1. All basic examinations shall be carried out using the multi-choice question format and essay questions as specified below. The incorrect alternatives shall seem equally plausible to anyone ignorant of the subject. All of the alternatives shall be clearly related to the question and of similar vocabulary, grammatical construction and length. In numerical questions, the incorrect answers shall correspond to procedural errors such as corrections applied in the wrong sense or incorrect unit conversions: they shall not be mere random numbers.
- 1.2. Each multi-choice question shall have three alternative answers of which only one shall be the correct answer and the candidate shall be allowed a time per module which is based upon a nominal average of 75 seconds per question.
- 1.3. Each essay question requires the preparation of a written answer and the candidate shall be allowed 20 minutes to answer each such question.
- 1.4. Suitable essay questions shall be drafted and evaluated using the knowledge syllabus in [Appendix I](#) Module 7.
- 1.5. Each question will have a model answer drafted for it, which will also include any known alternative answers that may be relevant for other subdivisions.
- 1.6. The model answer will also be broken down into a list of the important points known as Key Points.
- 1.7. The pass mark for each module and sub-module multi-choice part of the examination is 75 %.
- 1.8. The pass mark for each essay question is 75 % in that the candidates answer shall contain 75 % of the required key points addressed by the question and no significant error related to any required key point.
- 1.9. If either the multi-choice part only or the essay part only is failed, then it is only necessary to retake the multi-choice or essay part, as appropriate.
- 1.10. Penalty marking systems shall not be used to determine whether a candidate has passed.
- 1.11. An examination in a module may not be retaken earlier than 90 days following the date of a failed examination in that module, except in the case of a maintenance training organisation approved in accordance with [Annex IV \(Part-147\)](#) which delivers a course of retraining tailored to the failed subjects in the particular module; the failed module may be retaken after 30 days.
- 1.12. Basic knowledge examinations with a maximum allowed time of more than 90 or more than 180 minutes may be split in two or three partial exams respectively.

Each partial exam shall:

- (a) be complementary to the other partial exam or exams taken by the candidate, ensuring that the combination of partial exams meets the examination requirements for the subject module;
- (b) be of similar allowed time;
- (c) be passed with 75 % or more of the questions answered correctly;
- (d) contain a number of questions that is multiple of four;

- (e) be listed on the same certificate of recognition issued after the last partial exam has been successfully passed. That certificate of recognition shall list the dates and the results of the partial exams – without averaging the results;
- (f) be taken within the same organisation, following the normal examination provisions for retaking failed exams.

1.13. The maximum number of attempts for each examination is three in a 12-month period.

The applicant shall provide in a written statement to the approved maintenance training organisation or the competent authority to which they apply for an examination, the number, and dates of attempts during the 12 months preceding the examination, and the organisation or the competent authority where those attempts took place. The approved maintenance training organisation or the competent authority is responsible for checking the number of attempts within the applicable time frames.

1.14 While it is accepted that the subject matter of the questions may be the same, the questions used as part of the MBT learning programme shall not be used in examinations.

2. Number of questions per module

Regulation (EU) 2023/989

2.1. MODULE 1 – MATHEMATICS

Category A: 16 multiple-choice, no essay questions.

Time allowed: 20 minutes.

Category B1, B2, B2L and B3: 32 multiple-choice, no essay questions.

Time allowed: 40 minutes.

2.2. MODULE 2 – PHYSICS

Category A and B3: 32 multiple-choice, no essay questions.

Time allowed: 40 minutes.

Category B1, B2 and B2L: 52 multiple-choice, no essay questions.

Time allowed: 65 minutes.

2.3. MODULE 3 – ELECTRICAL FUNDAMENTALS

Category A: 20 multiple-choice, no essay questions.

Time allowed: 25 minutes.

Category B3: 24 multiple-choice, no essay questions.

Time allowed: 30 minutes.

Category B1, B2 and B2L: 52 multiple-choice, no essay questions.

Time allowed: 65 minutes.

2.4. MODULE 4 —ELECTRONICS FUNDAMENTALS

Category B1 and B3: 20 multiple-choice, no essay questions.

Time allowed: 25 minutes.

Category B2 and B2L: 40 multiple-choice, no essay questions.

Time allowed: 50 minutes.

2.5. MODULE 5 – DIGITAL TECHNIQUES/ELECTRONIC INSTRUMENT SYSTEMS

Category A and B3: 20 multiple-choice, no essay questions.

Time allowed: 25 minutes.

Category B1: 40 multiple-choice, no essay questions.

Time allowed: 50 minutes.

Category B2 and B2L: 72 multiple-choice, no essay questions.

Time allowed: 90 minutes.

2.6. MODULE 6 – MATERIALS AND HARDWARE

Category A: 52 multiple-choice, no essay questions.

Time allowed: 65 minutes.

Category B1 and B3: 80 multiple-choice, no essay questions.

Time allowed: 100 minutes.

Category B2 and B2L: 60 multiple-choice, no essay questions.

Time allowed: 75 minutes.

2.7. MODULE 7 – MAINTENANCE PRACTICES

Category A: 76 multiple-choice and 2 essay questions.

Time allowed: 95 minutes plus 40 minutes.

Category B1 and B3: 80 multiple-choice and 2 essay questions.

Time allowed: 100 minutes plus 40 minutes.

Category B2 and B2L: 60 multiple-choice and 2 essay questions.

Time allowed: 75 minutes plus 40 minutes.

2.8. MODULE 8 – BASIC AERODYNAMICS

Category A, B3, B1, B2 and B2L: 24 multiple-choice, no essay questions.

Time allowed: 30 minutes.

2.9. MODULE 9 – HUMAN FACTORS

Category A, B1, B3, B2 and B2L: 28 multiple-choice, no essay questions.

Time allowed: 35 minutes.

2.10. MODULE 10 – AVIATION LEGISLATION

Category A: 32 multiple-choice, no essay questions.

Time allowed: 40 minutes.

Category B1, B3, B2 and B2L: 44 multiple-choice, no essay questions.

Time allowed: 55 minutes.

2.11. MODULE 11 AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS

Category A1: 108 multiple-choice, no essay questions.

Time allowed: 135 minutes.

Category A2: 72 multiple-choice, no essay questions.

Time allowed: 90 minutes.

Category B1.1: 140 multiple-choice, no essay questions.

Time allowed: 175 minutes.

Category B1.2: 100 multiple-choice, no essay questions.

Time allowed: 125 minutes.

Category B3: 60 multiple-choice, no essay questions.

Time allowed: 75 minutes.

2.12. MODULE 12 – HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS:

Category A: 100 multiple-choice, no essay questions.

Time allowed: 125 minutes.

Category B1.3 and B1.4: 128 multiple-choice, no essay questions.

Time allowed: 160 minutes.

2.13. MODULE 13 – AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS

Category B2: 188 multiple-choice, no essay questions.

Time allowed: 235 minutes.

Category B2L:

System rating	Number of multiple-choice questions	Time allowed (minutes)
Basic requirements (Submodules 13.1, 13.2, 13.5 and 13.9)	32	40
COM/NAV (Submodule 13.4(a))	24	30
INSTRUMENTS (Submodule 13.8)	20	25
AUTOFLIGHT (Submodules 13.3 and 13.7)	28	35
SURVEILLANCE (Submodule 13.4(b))	20	25
AIRFRAME SYSTEMS (Submodules 13.11 to 13.19)	52	65

2.14. MODULE 14 — PROPULSION

Category B2 and B2L: 32 multiple-choice, no essay questions.

Time allowed: 40 minutes.

NOTE: The B2L examination for Module 14 is only applicable to the 'Instruments' and 'Airframe Systems' ratings.

2.15. MODULE 15 – GAS TURBINE ENGINE

Category A1 and A3: 60 multiple-choice, no essay questions.

Time allowed: 75 minutes.

Category B1.1 and B1.3: 92 multiple-choice, no essay questions.

Time allowed: 115 minutes.

2.16. MODULE 16 – PISTON ENGINE

Category A2 and A4: 52 multiple-choice, no essay questions.

Time allowed: 65 minutes.

Category B3, B1.2 and B1.4: 76 multiple-choice, no essay questions.

Time allowed: 95 minutes.

2.17. MODULE 17 – PROPELLER

Category A1 and A2: 20 multiple-choice, no essay questions.

Time allowed: 25 minutes.

Category B3, B1.1 and B1.2: 32 multiple-choice, no essay questions.

Time allowed: 40 minutes.

AMC1 Appendix II — Basic examination standard (except for category L licence)

ED Decision 2023/019/R

The tables below show the acceptable number of questions for the submodules. Justified deviations from these values are also acceptable, provided the sum of the questions complies with the total number of questions for a given module.

MODULE 1 — MATHEMATICS

MODULE 1 — MATHEMATICS	Nr of questions	
	A	B1 B2 B2L B3
Total number for the module:	16	32
1.1 Arithmetic	6	6
1.2 Algebra:		
(a) Simple algebraic expressions;	5	4
(b) Equations.	—	12
1.3 Geometry:		
(a) Simple geometrical constructions;	—	3
(b) Graphical representation;	5	4
(c) Trigonometry.	—	3

MODULE 2 — PHYSICS

MODULE 2 — PHYSICS	Nr of questions	
	A B3	B1 B2 B2L
Total number for the module:	32	52
2.1 Matter	4	5
2.2 Mechanics		
2.2.1 Statics	6	7
2.2.2 Kinetics	6	7
2.2.3 Dynamics:		
(a) Mass, force, and energy;	4	5
(b) Momentum and conservation of momentum.	4	4
2.2.4 Fluid dynamics:		
(a) Gravity and density;	2	2
(b) Viscosity; compressibility on fluids; static, dynamic, and total pressure.	2	3
2.3 Thermodynamics:		
(a) Temperature;	2	2
(b) Heat.	2	8
2.4 Optics (light)	—	5
2.5 Wave motion and sound	—	4

MODULE 3 — ELECTRICS FUNDAMENTALS

MODULE 3 — ELECTRICAL FUNDAMENTALS	Nr of questions		
	A	B1 B2 B2L	B3
Total number for the module:	20	52	24
3.1 Electron theory	2	2	2
3.2 Static electricity and conduction	3	3	2
3.3 Electrical terminology	3	2	2
3.4 Generation of electricity	3	2	2
3.5 Sources of DC electricity	3	3	3
3.6 DC circuits	1	2	1
3.7 Resistance/resistor:			
(a) Resistance;	—	3	1
(b) Resistors.	—	2	—
3.8 Power	—	3	1
3.9 Capacitance/capacitor	—	4	1
3.10 Magnetism:			
(a) Theory of magnetism;	—	3	1
(b) Magnetomotive force.	—	1	1
3.11 Inductance/inductor	—	4	1
3.12 DC motor/generator theory	—	3	1
3.13 AC theory	5	3	1

3.14 Resistive (R), capacitive (C) and inductive (L) circuits	—	3	1
3.15 Transformers	—	3	1
3.16 Filters	—	1	—
3.17 AC generators	—	3	1
3.18 AC motors	—	2	1

MODULE 4 — ELECTRONICS FUNDAMENTALS

MODULE 4 — ELECTRONICS FUNDAMENTALS	Nr of questions		
	A	B1 B3	B2 B2L
Total number for the module:	—	20	40
4.1 Semiconductors			
4.1.1 Diodes:			
(a) Description and characteristics;	—	8	8
(b) Operation and function.	—	—	7
4.1.2 Transistors:			
(a) Description and characteristics;	—	4	4
(b) Construction and operation.	—	—	7
4.1.3 Integrated circuits:			
(a) Basic description and operation;	—	3	2
(b) Description and operation.	—	—	4
4.2 Printed circuit boards	—	2	3
4.3 Servomechanisms:			
(a) Principles;	—	3	2
(b) Construction, operation, and use.	—	—	3

MODULE 5 — DIGITAL TECHNIQUES / ELECTRONIC INSTRUMENT SYSTEMS

MODULE 5 — DIGITAL TECHNIQUES / ELECTRONIC INSTRUMENT SYSTEMS	Nr of questions			
	A	B3	B1	B2 B2L
Total number for the module:	20	20	40	72
5.1 Electronic instrument systems	4	4	4	4
5.2 Numbering systems	—	—	3	5
5.3 Data conversion	—	—	3	4
5.4 Data buses	—	—	3	5
5.5 Logic circuits:				
(a) Identification and applications;	—	—	3	4
(b) Interpretation of logic diagrams.	—	—	—	4
5.6 Basic computer structure:				
(a) Computer terminology and technology;	6	2	4	2
(b) Computer operation.	—	—	—	6
5.7 Microprocessors	—	—	—	4
5.8 Integrated circuits	—	—	—	5
5.9 Multiplexing	—	—	—	4
5.10 Fibre optics	—	—	3	3

5.11 Electronic displays	2	2	2	4
5.12 Electrostatic sensitive devices	6	6	4	5
5.13 Software management control	—	1	3	3
5.14 Electromagnetic environment	—	1	3	4
5.15 Typical electronic/digital aircraft systems	2	4	5	6

MODULE 6 — MATERIALS AND HARDWARE

MODULE 6 — MATERIALS AND HARDWARE	Nr of questions		
	A	B1 B3	B2 B2L
Total number for the module:	52	80	60
6.1 Aircraft materials — ferrous:			
(a) Alloy steels used in aircraft;	3	3	3
(b) Testing of ferrous materials;	—	2	1
(c) Repair and inspection procedures.	—	2	1
6.2 Aircraft materials — non-ferrous:			
(a) Characteristics;	3	4	3
(b) Testing of non-ferrous materials;	—	3	2
(c) Repair and inspection procedures.	—	2	1
6.3 Aircraft materials — composite and non-metallic			
6.3.1 Composite and non-metallic other than wood and fabric:			
(a) Characteristics;	2	4	3
(b) Detection of defects;	2	4	—
(c) Repairs and inspection procedures.	—	2	2
6.3.2 Wooden structures	3	4	—
6.3.3 Fabric covering	—	4	—
6.4 Corrosion:			
(a) Chemical fundamentals;	3	3	3
(b) Types of corrosion.	4	5	3
6.5 Fasteners			
6.5.1 Screw threads	4	4	3
6.5.2 Bolts, studs, and screws	6	6	5
6.5.3 Locking devices	2	2	2
6.5.4 Aircraft rivets	2	3	2
6.6 Pipes and unions:			
(a) Identification;	1	1	1
(b) Standard unions.	2	2	2
6.7 Springs	—	1	1
6.8 Bearings	2	4	3
6.9 Transmissions	3	4	4
6.10 Control cables	5	5	4
6.11 Electrical cables and connectors	5	6	11

MODULE 7 — MAINTENANCE PRACTICES

MODULE 7 — MAINTENANCE PRACTICES	Nr of questions		
	A	B1 B3	B2 B2L
Total number for the module:	76	80	60
7.1 Safety precautions — aircraft and workshop	4	4	4
7.2 Workshop practices	4	4	4
7.3 Tools	6	6	6
7.4 (Reserved)	—	—	—
7.5 Engineering drawings, diagrams, and standards	6	6	6
7.6 Fits and clearances	5	5	5
7.7 Electrical wiring interconnection system (EWIS)	4	4	8
7.8 Riveting	4	3	—
7.9 Pipes and hoses	4	3	—
7.10 Springs	1	1	—
7.11 Bearings	4	3	—
7.12 Transmissions	3	3	—
7.13 Control cables	4	3	—
7.14 Material handling			
7.14.1 Sheet metal	—	2	—
7.14.2 Composite and non-metallic	—	2	—
7.14.3 Additive manufacturing	2	4	2
7.15 (Reserved)	—	—	—
7.16 Aircraft weight and balance:			
(a) Centre-of-gravity calculation;	—	2	2
(b) Aircraft weighing.	—	1	—
7.17 Aircraft handling and storage	7	5	6
7.18 Disassembly, inspection, repair, and assembly techniques:			
(a) Types of defects and visual inspection techniques;	2	2	2
(b) General repair methods — structural repair manual;	—	2	—
(c) Non-destructive inspection techniques;	—	1	1
(d) Disassembly and reassembly techniques;	2	1	1
(e) Troubleshooting techniques.	—	1	1
7.19 Abnormal events:			
(a) Inspections following lightning strikes and HIRF penetration;	2	1	2
(b) Inspections following abnormal events such as heavy landing and flight through turbulence.	2	1	—
7.20 Maintenance procedures	6	6	6
7.21 Documentation and communication	4	4	4

MODULE 8 — BASIC AERODYNAMICS

MODULE 8 — BASIC AERODYNAMICS	Nr of questions	
	A B3	B1 B2 B2L
Total number for the module:	24	24
8.1 Physics of the atmosphere	2	2
International Standard Atmosphere (ISA), application to aerodynamics.		
8.2 Aerodynamics	9	9
8.3 Theory of flight	7	7
8.4 High-speed airflow	4	4
8.5 Flight stability and dynamics	2	2

MODULE 9 — HUMAN FACTORS

MODULE 9 — HUMAN FACTORS	Nr of questions
	ALL
Total number for the module:	28
9.1 General	3
9.2 Human performance and limitations	3
9.3 Social psychology	2
9.4 Factors that affect human performance	3
9.5 Physical environment	2
9.6 Tasks	2
9.7 Communication	3
9.8 Human error	4
9.9 Safety management	2
9.10 The 'Dirty Dozen' and risk-mitigation	4

MODULE 10 — AVIATION LEGISLATION

MODULE 10 — AVIATION LEGISLATION	Nr of questions	
	A	B1 B2 B2L B3
Total number for the module:	32	44
10.1 Regulatory framework	5	5
10.2 Certifying staff — maintenance	7	7
10.3 Approved maintenance organisations	6	6
10.4 Independent certifying staff	-	4
10.5 Air operations	4	4
10.6 Certification of aircraft, parts, and appliances	1	4
10.7 Continuing airworthiness	6	7
10.8 Oversight principles in continuing airworthiness	2	3

MODULE 10 — AVIATION LEGISLATION	Nr of questions	
	A	B1 B2 B2L B3
10.9 Maintenance and certification beyond the current EU regulations (if not superseded by EU requirements)	—	3
10.10 Cybersecurity in aviation maintenance	1	1

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions				
	A1	A2	B1.1	B1.2	B3
Total number for the module:	108	72	140	100	60
11.1 Theory of flight:					
(a) Aeroplane aerodynamics and flight controls;	2	2	2	2	2
(b) Aeroplane, other aerodynamic devices.	2	2	2	2	2
11.2 Airframe structures (ATA 51) :					
(a) General concepts;	3	3	4	3	2
(b) Airworthiness requirements for structural strength;	3	3	3	3	1
(c) Construction methods.	1	1	3	2	1
11.3 Airframe structures — aeroplanes					
11.3.1 Fuselage, doors, windows (ATA 52/53/56) :	1	1	2	2	2
(a) Construction principles;					
(b) Airborne towing devices;	1	1	1	1	1
(c) Doors.	1	1	1	1	-
11.3.2 Wings (ATA 57)	2	2	3	3	2
11.3.3 Stabilisers (ATA 55)	1	1	2	2	1
11.3.4 Flight control surfaces (ATA 55/57)	1	1	2	2	1
11.3.5 Nacelles/pylons (ATA 54)	1	1	2	2	1
11.4 Air conditioning and cabin pressurisation (ATA 21) :					
(a) Pressurisation;	2	2	3	1	—
(b) Air supply;	3	—	3	—	—
(c) Air conditioning;	3	—	3	—	—
(d) Safety and warning devices;	2	1	2	2	—
(e) Heating and ventilation system.	—	1	—	2	1
11.5 Instruments / avionics systems					
11.5.1 Instrument systems (ATA 31)	2	2	4	4	3
11.5.2 Avionics systems	3	2	5	4	4
Fundamentals of system layouts and operation of:					

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions				
	A1	A2	B1.1	B1.2	B3
— Autoflight (ATA 22); — Communications (ATA 23); — Navigation systems (ATA 34).					
11.6 Electrical power (ATA 24)	4	3	5	5	4
11.7 Equipment and furnishings (ATA 25)					
(a) Emergency equipment;	4	2	4	3	2
(b) Cabin and cargo layout.	3	3	3	3	-
11.8 Fire protection (ATA 26)					
(a) Fire and smoke detection system and fire-extinguishing systems;	3	2	4	3	—
(b) Portable fire extinguisher.	1	1	1	1	1
11.9 Flight controls (ATA 27)					
(a) Primary and secondary flight controls;	3	2	4	4	3
(b) Actuation and protection;	3	—	3	—	—
(c) System operation;	3	—	3	—	—
(d) Balancing and rigging.	1	1	3	3	2
11.10 Fuel systems (ATA 28/47)					
(a) Systems layout;	2	2	3	3	2
(b) Fuel handling;	2	2	2	2	1
(c) Indications and warnings;	1	1	2	2	1
(d) Special systems;	1	—	1	—	—
(e) Balancing.	2	—	2	—	—
11.11 Hydraulic power (ATA 29)					
(a) System description;	1	1	3	3	2
(b) System operation (1);	1	1	3	2	1
(c) System operation (2).	2	—	2	—	—
11.12 Ice and rain protection (ATA 30)					
(a) Principles;	1	1	2	2	1
(b) De-icing;	1	1	2	2	1
(c) Anti-icing;	1	—	2	—	—
(d) Wipers;	1	1	2	2	1
(e) Rain-repellent systems.	2	—	2	—	—
11.13 Landing gear (ATA 32)					
(a) Description;	3	3	4	3	1
(b) System operation;	3	3	4	2	1
(c) Air-ground sensing;	1	—	1	—	—

MODULE 11 — AEROPLANE AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions				
	A1	A2	B1.1	B1.2	B3
(d) Tail protection.	1	1	1	1	1
11.14 Lights (ATA 33)	2	2	3	3	2
11.15 Oxygen (ATA 35)	3	3	4	4	3
11.16 Pneumatic/vacuum (ATA 36)					
(a) Systems;	3	3	3	3	2
(b) Pumps.	3	3	3	3	2
11.17 Water/waste (ATA 38)					
(a) Systems;	2	2	2	2	1
(b) Corrosion.	1	1	1	1	1
11.18 Onboard maintenance systems (ATA 45)	3	—	3	—	—
11.19 Integrated modular avionics (IMA) (ATA 42)					
(a) Overall system description and theory;	1	—	1	—	—
(b) Typical system layouts.	1	—	1	—	—
11.20 Cabin systems (ATA 44)	2	—	2	—	—
11.21 Information systems (ATA 46)	2	—	2	—	—

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions	
	A3 A4	B1.3 B1.4
Total number for the module:	100	128
12.1 Theory of flight — rotary wing aerodynamics	6	9
12.2 Flight control systems (ATA 67)	9	9
12.3 Blade tracking and vibration analysis (ATA 18)	6	9
12.4 Transmission	3	6
12.5 Airframe structures (ATA 51)		
(a) General concept;	5	6
(b) Construction methods of the principal elements.	5	7
12.6 Air conditioning (ATA 21)	—	—
12.6.1 Air supply	1	2
12.6.2 Air conditioning	3	5
12.7 Instruments / avionics systems		
12.7.1 Instrument systems (ATA 31)	6	9
12.7.2 Avionics systems	5	7
Fundamentals of system layouts and operation of:		
— Autoflight (ATA 22);		
— Communications (ATA 23);		
— Navigation systems (ATA 34).		
12.8 Electrical power (ATA 24)	8	10

MODULE 12 — HELICOPTER AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions	
	A3 A4	B1.3 B1.4
12.9 Equipment and furnishings (ATA 25)		
(a) Emergency equipment: Seats, harnesses, and belts; Lifting systems.	2	3
(b) Emergency flotation systems: Cabin layout, cargo retention; Equipment layout; Cabin furnishing installation.	3	3
12.10 Fire protection (ATA 26)		
(a) Fire and smoke detection systems and fire-extinguishing systems;	3	4
(b) Portable fire extinguishers.	1	1
12.11 Fuel systems (ATA 28)	7	8
12.12 Hydraulic power (ATA 29)	8	8
12.13 Ice and rain protection (ATA 30)	4	4
12.14 Landing gear (ATA 32)		
(a) System description and operation;	3	4
(b) Sensors.	3	3
12.15 Lights (ATA 33)	3	4
12.16 (Reserved)		
12.17 Integrated modular avionics (IMA) (ATA 42)		
(a) Overall system description and theory;	1	1
(b) Typical system layouts.	1	1
12.18 Onboard maintenance systems (ATA 45)	2	3
Central maintenance computers; Data-loading system; Electronic library system.		
12.19 Information systems (ATA 46)	2	2

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS

C/N: Communication and Navigation; **Ins.:** instruments; **A/F:** Autoflight; **Sur.:** Surveillance; **A/S:** Airframe and Systems

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
Total number for the module:	188	32	24	20	28	20	50
13.1 Theory of flight							
(a) Aeroplane aerodynamics and flight controls;	3	3	—	—	—	—	—
(b) Rotary wing aerodynamics.	1	1	—	—	—	—	—
13.2 Structures — general concepts (ATA 51)							
(a) General concepts;	4	4	—	—	—	—	—
(b) Fundamentals of structural systems.	4	4	—	—	—	—	—
13.3 Autoflight (ATA 22)							
(a) Fundamentals of automatic flight control;	16	—	—	—	8	—	—
(b) Autothrottle systems and automatic landing systems.	8	—	—	—	8	—	—
13.4 Communication/navigation (ATA 23/34)							
(a) Fundamentals of communication and navigation systems;	24	—	24	—	—	—	—
(b) Fundamentals of aircraft surveillance systems.	3	—	—	—	—	20	—
13.5 Electrical power (ATA 24)	13	13	—	—	—	—	—
13.6 Equipment and furnishings (ATA 25)	5	—	—	—	—	—	—
13.7 Flight controls							
(a) Primary and secondary flight controls (ATA 27);	4	—	—	—	3	—	—
(b) Actuation and protection;	4	—	—	—	3	—	—
(c) System operation;	2	—	—	—	3	—	—
(d) Rotorcraft flight controls (ATA 67).	2	—	—	—	3	—	—
13.8 Instruments (ATA 31)	20	—	—	20	—	—	—
13.9 Lights (ATA 33)	7	7	—	—	—	—	—
13.10 Onboard maintenance systems (ATA 45)	5	—	—	—	—	—	—
13.11 Air conditioning and cabin pressurisation (ATA 21)							
(a) Pressurisation;	2	—	—	—	—	—	2
(b) Air supply;	2	—	—	—	—	—	2
(c) Air conditioning;	2	—	—	—	—	—	2
(d) Safety and warning devices.	2	—	—	—	—	—	2
13.12 Fire protection (ATA 26)							
(a) Fire and smoke detection system and fire-extinguishing systems;	2	—	—	—	—	—	2
(b) Portable fire extinguisher.	1	—	—	—	—	—	1
13.13 Fuel systems (ATA 28, ATA 47)							
(a) System layout;	2	—	—	—	—	—	2
(b) Fuel handling;	2	—	—	—	—	—	2
(c) Indications and warnings;	2	—	—	—	—	—	2
(d) Special systems;	2	—	—	—	—	—	2
(e) Balancing.	1	—	—	—	—	—	1
13.14 Hydraulic power (ATA 29)							

MODULE 13 — AIRCRAFT AERODYNAMICS, STRUCTURES AND SYSTEMS	Nr of questions						
	B2	B2L Basic	B2L C/N	B2L Ins.	B2L A/F	B2L Sur.	B2L A/S
(a) System layout;	1	—	—	—	—	—	1
(b) System operation (1);	5	—	—	—	—	—	4
(c) System operation (2).	5	—	—	—	—	—	4
13.15 Ice and rain protection (ATA 30)							
(a) Principles;	1	—	—	—	—	—	1
(b) De-icing;	2	—	—	—	—	—	2
(c) Anti-icing;	1	—	—	—	—	—	1
(d) Wiper systems;	1	—	—	—	—	—	1
(e) Rain repellent.	1	—	—	—	—	—	1
13.16 Landing gear (ATA 32)							
(a) Description;	1	—	—	—	—	—	1
(b) System;	3	—	—	—	—	—	3
(c) Air–ground sensing.	3	—	—	—	—	—	3
13.17 Oxygen (ATA 35)	2	—	—	—	—	—	—
13.18 Pneumatic/vacuum (ATA 36)	6	—	—	—	—	—	6
13.19 Water/waste (ATA 38)	2	—	—	—	—	—	2
13.20 Integrated modular avionics (IMA) (ATA 42)							
(a) Overall system description and theory;	2	—	—	—	—	—	—
(b) Typical system layouts.	1	—	—	—	—	—	—
13.21 Cabin systems (ATA 44)	3	—	—	—	—	—	—
13.22 Information systems (ATA 46)	3	—	—	—	—	—	—

MODULE 14 — PROPULSION

MODULE 14 — PROPULSION	Nr of questions
	B2 B2L Instruments B2L Airframe and Systems
Total number for the module:	32
14.1 Engines	
(a) Turbine engines;	3
(b) Auxiliary power units (APUs);	4
(c) Piston engines;	2
(d) Electric and hybrid engines;	4
(e) Engine control.	3
14.2 Electric/electronic engine indication systems	10
14.3 Propeller systems	2
14.4 Starting and ignition systems	4

MODULE 15 — GAS-TURBINE ENGINE

MODULE 15 — GAS-TURBINE ENGINE	Nr of questions	
	A1 A3	B1.1 B1.3
Total number for the module:	60	92
15.1 Fundamentals	4	4
15.2 Engine performance	—	6
15.3 Inlet	3	4
15.4 Compressors	5	7
15.5 Combustion section	3	3
15.6 Turbine section	5	5
15.7 Exhaust	4	4
15.8 Bearings and seals	—	3
15.9 Lubricants and fuels	3	4
15.10 Lubrication systems	3	4
15.11 Fuel systems	4	5
15.12 Air systems	3	3
15.13 Starting and ignition systems	3	4
15.14 Engine indication systems	5	7
15.15 Alternate turbine constructions	—	2
15.16 Turboprop engines	3	5
15.17 Turboshift engines	2	3
15.18 Auxiliary power units (APUs)	2	3
15.19 Power plant installation	2	3
15.20 Fire protection systems	2	3
15.21 Engine monitoring and ground operation	4	7
15.22 Engine storage and preservation	—	3

MODULE 16 — PISTON ENGINE

MODULE 16 — PISTON ENGINE	Nr of questions	
	A2 A4	B1.2 B1.4 B3
Total number for the module:	52	76
16.1 Fundamentals of piston engines	5	5
16.2 Engine performance	3	5
16.3 Engine construction	7	8
16.4 Engine fuel systems	—	—
16.4.1 Carburettors	3	4
16.4.2 Fuel injection systems	2	4
16.4.3 Electronic engine control	2	4
16.5 Starting and ignition systems	5	5
16.6 Induction, exhaust, and cooling systems	3	4
16.7 Supercharging/turbocharging	4	6
16.8 Lubricants and fuels	2	5
16.9 Lubrication systems	3	4
16.10 Engine indication systems	6	7
16.11 Power plant installation	3	3
16.12 Engine monitoring and ground operation	3	5
16.13 Engine storage and preservation	—	3
16.14 Alternative piston-engine constructions	1	4

MODULE 17 — PROPELLER

MODULE 17 — PROPELLER	Nr of questions	
	A1 A2	B1.1 B1.2 B3
Total number for the module:	20	32
17.1 Fundamentals of propellers	5	8
17.2 Propeller construction	4	5
17.3 Propeller pitch control	4	6
17.4 Propeller synchronising	—	2
17.5 Propeller ice protection	2	3
17.6 Propeller maintenance	3	6
17.7 Propeller storage and preservation	2	2